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Novel benzimidazole–oxadiazole hybrid molecules as promising antimicrobial agents†

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In the present study, we describe the design and expeditious synthesis of novel 2-aryl-5-(3-aryl-[1,2,4]-oxadiazol-5-yl)-1-methyl-1*H*-benzo[d]imidazole hybrid molecules as promising antimicrobial agents. The core moiety 2-aryl-ethyl-1*H*-benzo[d]imidazole-5-carboxylate was efficiently synthesized by a rapid 'one pot' nitro reductive cyclization reaction using sodium dithionite as reagent. All the compounds were screened for their antimicrobial activities and the active compounds were screened for their anti-tubercular activity against *Mycobacterium tuberculosis* H37Rv strain by the Microplate Alamar Blue Assay method. Compounds **8k**, **8n**, **8p** and **8r** exhibited potent anti-tubercular activity with MIC of 1.6 $\mu\text{g mL}^{-1}$, which is two times more potent than the standard drugs pyrazinamide and ciprofloxacin and fourfold superior to streptomycin and isoniazid. Further, potent compounds were tested for their preliminary toxicity by hemolytic assay, where the compounds remain nontoxic even at higher concentration and showed good selectivity index. The investigation of cytotoxicity against normal embryonic kidney cell line HEK 293 by MTT assay showed IC_{50} value of more than 355 $\mu\text{g mL}^{-1}$ for all the tested potent compounds. From the screening study, **8k**, **8n**, **8p** and **8r** emerged as strong candidates with potent antimicrobial activities and good ADME parameters. Attempts were also made to establish the structural activity relationships among the tested compounds.

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1. Introduction

Antibiotics are the most important weapons in the fight against microbial infections and have enormously benefited the health-related quality of human life since their introduction. However, over the past few decades, these health benefits are under threat as many commonly used antibiotics are becoming less sensitive and less effective against certain illnesses, not only because of toxic reactions, but also due to the emergence of drug-resistant bacteria. After AIDS, tuberculosis (TB) caused by *Mycobacterium tuberculosis* is the world's second most serious and fatal chronic infectious disease.¹ People with weak immune systems, for example, HIV positive, diabetes,² cancer³ and transplant⁴ patients, are more susceptible to TB infection. The unique

structure and chemical composition of the mycobacterial cell wall prevent successful entry of drugs which renders most antibiotics ineffective making TB treatment very difficult. Furthermore, this problem has become more critical with the emergence of multi-drug-resistant and extensively drug-resistant strains of *M. tuberculosis* against both the first line as well as second line drugs.^{5–7} But no such new effective TB drugs have been launched in the market for the past few decades. It is also necessary that effective therapeutic agents should possess good safety profiles, i.e. good selectivity index and less toxicity. But there are many reports found in the literature where the most potent compounds are not effective because of their toxicity. Also, potency and selectivity are extremely sensitive to minor changes in the chemical structure of the drug.^{8–12} Hence, there is an urgent requirement of developing new antimycobacterial drugs with a unique structure, effective mode of action, with reduced toxicity and good selectivity compared to the existing drugs.

The benzimidazole nucleus is considered as 'Master Key Core' in many compounds aimed at different targets to bring out varied pharmacological properties. Benzimidazole chemistry has gained prominent importance with the discovery of *N*-ribosyldimethylbenzimidazole which serves as an axial ligand for cobalt in vitamin B₁₂.¹³ In the synthesis of benzimidazoles reduction of nitro group followed by cyclization is the main process which conventionally involves two steps. One pot reactions are gaining significant importance to conduct syntheses in a more sustainable

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fashion with an aim of minimizing reagent and solvent use, as well as to reduce the isolation steps.¹⁴ Benzimidazole and its derivatives represent one of the highly important biologically active classes of compounds possessing various biological activities including antiviral,¹⁵ anthelmintic,¹⁶ anticancer,^{17,18} antihistamine,¹⁹ antihypertensive,²⁰ antineoplastic,²¹ antiprotozoal,²² and anti-hepatitis B virus activities.²³ Only few reports are available of antitubercular activity of benzimidazoles.^{24–27} These developments sparked active research to explore the nucleus with varied activities. Optimization of substituents around the benzimidazole nucleus has resulted in many drugs like carbendazim (antifungal agent), albendazole, mebendazole (anti-helminthic agents), pimozide (anti-psychotic drug), omeprazole, pantoprazole (proton pump inhibitors) and many lead compounds are used in wide therapeutic areas.

Moreover, most of the drugs have 2nd and 5th position substitutions at the benzimidazole core. Substitutions at these two positions of the benzimidazole moiety were identified as essential for receptor activation and potency of receptor binding.²⁸ On the other hand, [1,2,4]-oxadiazole is another privileged scaffold finding great importance in medicinal chemistry research due to its synthetic applications such as anticonvulsant,²⁹ analgesic,³⁰ anti-inflammatory,³¹ and antitumor³²

applications. Also, [1,2,4]-oxadiazoles are reported for their selective inhibition of receptors such as tyrosinase inhibitors,³³ muscarinic,³⁴ histamine- H_3 ³⁵ and benzodiazepines.³⁶ In addition, significant numbers of drugs and drug candidates in clinical development are halogenated. Insertion of halogen atoms in hit or lead compounds predominantly causes development in their steric effects by increasing the ability of bulky atoms to occupy the binding sites of molecular targets and also increases the polar hydrophobicity.³⁷

Based on the biological profiles of the benzimidazoles and [1,2,4]-oxadiazole and also in continuation of our research work on synthesis of benzimidazoles by the 'one pot' nitro reductive cyclization method,^{38,39} we report the synthesis of a series of novel benzimidazoles with [1,2,4]-oxadiazole in a single molecular frame and their evaluation for antibacterial, antifungal and antimycobacterial activities (Fig. 1).

2. Results and discussion

2.1. Chemistry

Twenty new compounds were synthesized by a series of reactions as depicted in Schemes 1–3. The intermediates N^1 -

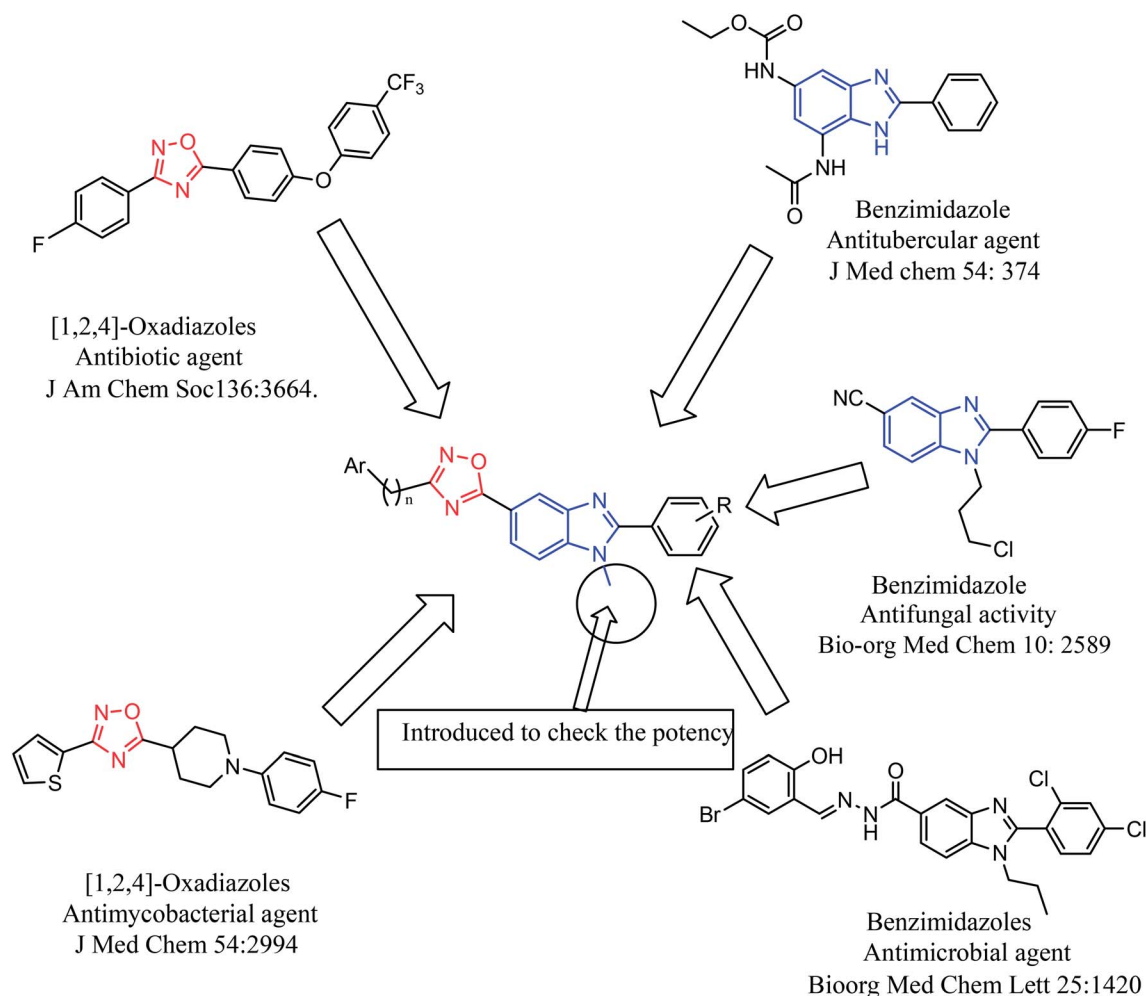
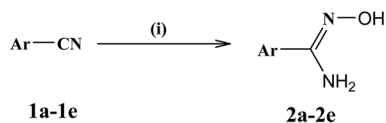


Fig. 1 Design strategy for the generation of benzimidazoles with [1,2,4]-oxadiazoles scaffolds.



Ar: 4-Cl-C₆H₄, 3-C₅H₄N, 2-Cl-C₆H₄, 3-Cl-C₆H₄-CH₂, 4-CH₃-SO₂-C₆H₄

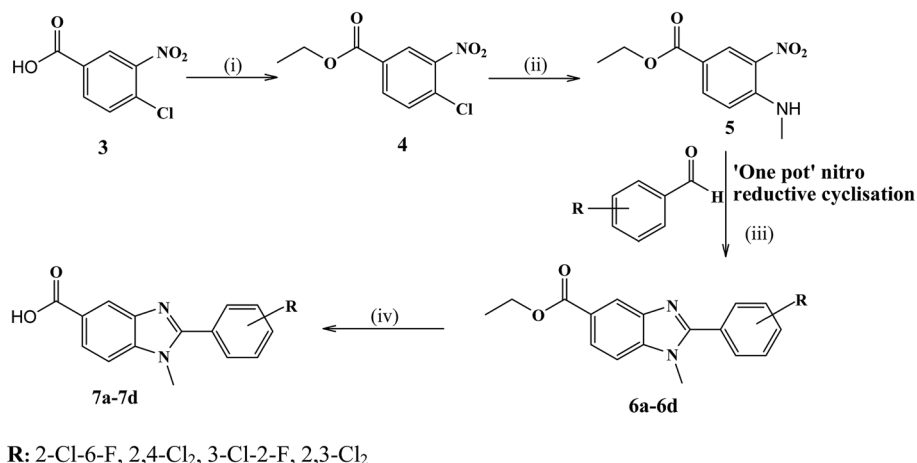
Scheme 1 Synthesis of *N*¹-hydroxyamidines **2a–e**. Reagents and conditions: (i) hydroxylamine hydrochloride, sodium bicarbonate, EtOH, RT, 6 h.

hydroxyamidines (**2a–e**) were synthesized by treating various aromatic/heterocyclic nitriles (**1a–e**) with hydroxylamine hydrochloride in ethanol using sodium bicarbonate. The intermediate ethyl 4-chloro-3-nitrobenzoate (**4**) was synthesized by esterification of commercially available 4-chloro-3-nitrobenzoic acid **3** according to a reported literature procedure.⁴⁰ The compound ethyl 3-nitro-4-(methylamino)benzoate **5** was obtained by the nucleophilic displacement reaction of compound **4** with methylamine in the presence of triethylamine as a base in THF solvent. The core nucleus ethyl 2-aryl-1-methyl-1*H*-benzo[*d*]imidazole-5-carboxylate (**6a–d**) was prepared by 'one-pot' nitro reductive cyclization of **5** with various halo substituted aromatic aldehydes using sodium dithionite reagent in dimethyl sulphoxide medium at 90 °C. This 'one pot' reductive synthesis of **6a–d** gave good yield (more than 90%) with high purity in very short reaction time as compared to conventional stepwise cyclization methods. Usually, in stepwise methods, the first step involves reduction of nitro group to an amine by hydrogenation or reduction reaction, followed by cyclization which requires strong acidic or oxidizing conditions. But this method involves several drawbacks, like harsh reaction conditions, use of expensive reagents, isolation of intermediates, tedious workup procedures, long reaction path and low yield. Hence, compared to stepwise methods, 'one pot' reductive cyclization is a simple, convenient, cheap and high yielding method which is much more superior to stepwise cyclization methods.³⁸ The compound 2-aryl-1-methyl-1*H*-benzo[*d*]imidazole-5-carboxylic

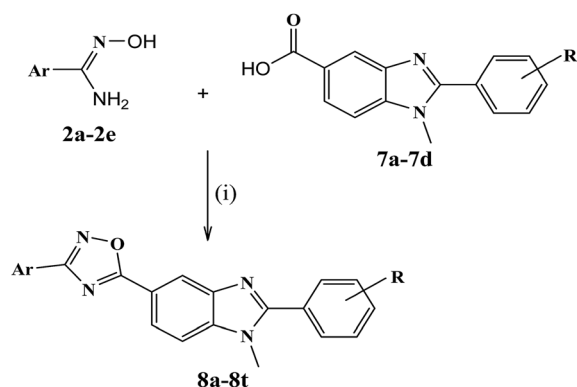
acid (**7**) was obtained by ester hydrolysis of 'one pot' intermediate **6a–d** by using NaOH in ethanol medium. Finally, intermediates *N*¹-hydroxyamidines (**2a–e**) undergo *O*-acylation with carboxylic acid group of intermediates **6a–d** followed by intramolecular cyclodehydration in the presence of coupling agent, TBTU and base *N*-methylmorpholine in dimethylformamide medium affording 2-aryl-5-(3-aryl-[1,2,4]-oxadiazol-5-yl)-1-methyl-1*H*-benzo[*d*]imidazoles (**8a–t**). Physical properties and structural features of target compounds **8a–t** are presented in Table 1.

The newly synthesized compounds were characterized by FT (ATR) IR, ¹H NMR, ¹³C NMR, ESI MS and elemental analyses. The formation of intermediates **2a–e** from corresponding aromatic nitriles was confirmed by the appearance of characteristic stretching vibrations for NH and OH groups. For the compound **2e**, NH and OH stretching vibrations were observed at 3342 and 3197 cm^{−1}. The characteristic stretching vibrations for SO₂ group were observed at 1382 cm^{−1} (asym) and 1123 cm^{−1} (sym). Also, in the ¹H NMR spectrum, free NH₂ signal appeared at δ 5.45 ppm and a distinct singlet for OH appeared at δ 8.93 ppm. Further, a singlet at δ 2.71 ppm indicated the presence of three protons of methylsulfonyl group. Two distinct doublets were observed at δ 7.47 and δ 7.60 ppm for four protons of methylsulfonyl phenyl ring, each integrating for two protons with coupling constants *J* = 8.4 Hz and 8.0 Hz respectively.

The formation of 'one pot' reductive cyclized core nucleus **6a–d** was confirmed by the absence of characteristic NH stretching at 3350 cm^{−1} of the methylamine compound **5** of the previous step. Further, by considering prototype **6a**, its ¹H NMR spectrum showed three protons of 2-chloro-6-fluorophenyl ring as a triplet at δ 7.52 ppm (coupling constant *J* = 8.4 Hz) and a multiplet in the region δ 7.42–7.46 ppm which clearly indicated the cyclization. In addition, the spectrum displayed a doublet centered at δ 7.62 ppm, a doublet of doublet in the region of δ 7.71–7.81 ppm and a singlet at δ 8.34 ppm due to three aromatic protons of benzimidazole ring. The –CH₃ and –CH₂ protons of ester group appeared as a triplet at δ 1.36 ppm and a quartet in the region δ 4.33–4.39 ppm. In its spectrum, the



Scheme 2 Synthesis of core nucleus ethyl 2-aryl-1-methyl-1*H*-benzo[*d*]imidazole-5-carboxylate **6a–d**. Reagents and conditions: (i) conc. H₂SO₄, EtOH, reflux, 6 h. (ii) Methylamine, triethylamine, THF, RT. (iii) Na₂S₂O₄, DMSO, 90 °C, 3 h. (iv) Sodium hydroxide, EtOH, reflux, 8 h.



Scheme 3 Synthesis of target compounds, 2-aryl-5-(3-aryl-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzimidazoles **8a–t**. Reagents and conditions: (i) TBTU, *N*-methyl morpholine, DMF, reflux, 8 h.

disappearance of a broad --NH peak in the downfield region strongly confirmed the formation of compound **6a**. Its formation was also evidenced by ESI MS spectral data which showed ($\text{M}^+ + \text{H}$) peak at m/z value of 333.1. In the next step, the FT (ATR) IR spectrum of compound **7a** showed the characteristic OH vibration at 3220 cm^{-1} which was absent in the spectrum of the corresponding previous ester **6a**. The shifting of carbonyl stretching frequency to a lower value of 1693 cm^{-1} from 1735 cm^{-1} gave strong evidence for the conversion of ester group to

acid. Further, its ^1H NMR spectrum displayed a broad OH signal at δ 12.82 ppm and also the absence of signals due to ester protons in the shielded region strongly supported the conversion of ester **6a** to acid **7a**. In its ESI MS spectrum, the presence of a deprotonated molecular ion peak at m/z value of 303.0 further confirmed its formation.

Formation of new target compounds **8a–t** was also characterized by spectral data and elemental analyses. In the FT (ATR) IR spectra, the absence of characteristic OH/NH vibrations strongly evidenced the formation of **8a–t**, which was found in the spectra of intermediates **2a–e**. Aromatic protons showed stretching vibrations around $3018\text{--}3080\text{ cm}^{-1}$. For the prototype compound **8a**, C–F and C–Cl stretching bands were observed at 1165 cm^{-1} and 764 cm^{-1} respectively. Further, the ^1H NMR spectrum of **8a** showed a characteristic singlet for N--CH_3 group at δ 3.67 ppm. Aromatic protons of 2-chloro-6-fluoro ring resonated as a multiplet in the region δ 7.42–7.46 ppm. The proton at C_7 of benzimidazole appeared as a doublet at δ 7.34 ppm with coupling constant $J = 7.6\text{ Hz}$, C_6 as a doublet of doublet in the region δ 8.17–8.19 ppm ($J = 8.4$ and 1.2 Hz) and C_4 as a broad singlet at 8.65 ppm. In its ^{13}C NMR spectrum, the signal due to carbonyl group was absent. N--CH_3 carbon of benzimidazole gave a signal in the region δ 30–34 ppm, along with all aromatic carbons in the region δ 110–160 ppm. Oxadiazole ring carbons showed signals in the downfield region δ 162–176 ppm, which provide an additional support

Table 1 Physical properties and structural features of target compounds **8a–t**

Compound	Ar	R	Mol. formula	Mol. weight	M. p. ($^{\circ}\text{C}$)
8a	4-Chlorophenyl	2-Chloro-6-fluoro	$\text{C}_{22}\text{H}_{13}\text{Cl}_2\text{FN}_4\text{O}$	439.27	212–214
8b	Pyridine-3-yl	2-Chloro-6-fluoro	$\text{C}_{21}\text{H}_{13}\text{ClFN}_5\text{O}$	405.81	222–224
8c	2-Chlorophenyl	2-Chloro-6-fluoro	$\text{C}_{22}\text{H}_{13}\text{Cl}_2\text{FN}_4\text{O}$	439.27	202–204
8d	3-Chlorobenzyl	2-Chloro-6-fluoro	$\text{C}_{23}\text{H}_{15}\text{Cl}_2\text{FN}_4\text{O}$	453.3	192–194
8e	4-(Methylsulfonyl)phenyl	2-Chloro-6-fluoro	$\text{C}_{23}\text{H}_{16}\text{ClFN}_4\text{O}_3\text{S}$	482.91	>225
8f	4-Chlorophenyl	2,4-Dichloro	$\text{C}_{23}\text{H}_{15}\text{Cl}_2\text{FN}_4\text{O}$	455.72	200–202
8g	Pyridine-3-yl	2,4-Dichloro	$\text{C}_{21}\text{H}_{13}\text{Cl}_2\text{N}_5\text{O}$	422.27	188–190
8h	2-Chlorophenyl	2,4-Dichloro	$\text{C}_{22}\text{H}_{13}\text{Cl}_3\text{N}_4\text{O}$	455.72	201–203
8i	3-Chlorobenzyl	2,4-Dichloro	$\text{C}_{23}\text{H}_{15}\text{Cl}_3\text{N}_4\text{O}$	469.75	198–200
8j	4-(Methylsulfonyl)phenyl	2,4-Dichloro	$\text{C}_{23}\text{H}_{16}\text{Cl}_2\text{N}_4\text{O}_3\text{S}$	499.37	184–186
8k	4-Chlorophenyl	3-Chloro-2-fluoro	$\text{C}_{22}\text{H}_{13}\text{Cl}_2\text{FN}_4\text{O}$	439.27	>225
8l	Pyridine-3-yl	3-Chloro-2-fluoro	$\text{C}_{21}\text{H}_{13}\text{ClFN}_5\text{O}$	405.81	186–189
8m	2-Chlorophenyl	3-Chloro-2-fluoro	$\text{C}_{22}\text{H}_{13}\text{Cl}_2\text{FN}_4\text{O}$	439.27	190–192
8n	3-Chlorobenzyl	3-Chloro-2-fluoro	$\text{C}_{23}\text{H}_{15}\text{Cl}_2\text{FN}_4\text{O}$	453.3	>225
8o	4-(Methylsulfonyl)phenyl	3-Chloro-2-fluoro	$\text{C}_{23}\text{H}_{16}\text{ClFN}_4\text{O}_3\text{S}$	482.91	>225
8p	4-Chlorophenyl	2,3-Dichloro	$\text{C}_{23}\text{H}_{15}\text{Cl}_2\text{FN}_4\text{O}$	455.72	194–196
8q	Pyridine-3-yl	2,3-Dichloro	$\text{C}_{21}\text{H}_{13}\text{Cl}_2\text{N}_5\text{O}$	422.27	>225
8r	2-Chlorophenyl	2,3-Dichloro	$\text{C}_{22}\text{H}_{13}\text{Cl}_3\text{N}_4\text{O}$	455.72	180–182
8s	3-Chlorobenzyl	2,3-Dichloro	$\text{C}_{23}\text{H}_{15}\text{Cl}_3\text{N}_4\text{O}$	469.75	172–174
8t	4-(Methylsulfonyl)phenyl	2,3-Dichloro	$\text{C}_{23}\text{H}_{16}\text{Cl}_2\text{N}_4\text{O}_3\text{S}$	499.37	>225

for the formation of the compound. Further, compound **8a** displayed a protonated peak at m/z value of 439.2 along with isotopic peaks at m/z values of 441.2 and 443.2 in its ESI MS spectrum which is in accordance with its molecular formula weight and evidences the formation of the target molecule. The spectral and elemental analysis data of all target compounds are presented in the experimental section.

2.2. Biological screening

2.2.1. Antibacterial activity. To investigate the antimicrobial potency, all the newly synthesized compounds **8a–t** were screened for their *in vitro* antibacterial activity against two representative Gram-positive bacterial species, *S. aureus* and *E. faecalis*, and two Gram-negative bacterial strains, *E. coli* and *K. pneumoniae*, by the broth dilution method⁴¹ using ciprofloxacin as reference drug in terms of minimum inhibitory concentration (MIC). Results are summarized in Table 2. From the obtained results, it has been noticed that the tested new compounds exhibited moderate to good inhibition in the range of $1.6 \mu\text{g mL}^{-1}$ to $50 \mu\text{g mL}^{-1}$. The compounds **8a** and **8k** showed the highest inhibition among the compounds in the series. Compound **8a** with 2-chloro-6-fluorophenyl substitution at the 2nd position of benzimidazole and *p*-chlorophenyl substitution at the 3rd position of oxadiazole ring at other extreme end showed MIC values of $1.6 \mu\text{g mL}^{-1}$ against *E. faecalis*, $3.125 \mu\text{g mL}^{-1}$ against *S. aureus*, $6.25 \mu\text{g mL}^{-1}$ against *K.*

pneumoniae and *E. coli*. Compound **8k** with 3-chloro-2-fluorophenyl substitution at the 2nd position of benzimidazole and *p*-chlorophenyl substitution at oxadiazole ring exhibited good inhibition with MIC value of $1.6 \mu\text{g mL}^{-1}$ against both Gram-positive bacterial strains, which is lower than that of the standard drug ciprofloxacin (MIC: $2.0 \mu\text{g mL}^{-1}$ against *S. aureus*, $6.25 \mu\text{g mL}^{-1}$ against *E. coli* and $12.5 \mu\text{g mL}^{-1}$ against *K. pneumoniae*). Surprisingly, both compounds **8a** and **8k** possess chlorofluorophenyl substitutions at benzimidazole ring. The higher activity could be attributed to the presence of chloro-fluoro groups; especially the presence of the highly active small 'fluorine' atom may impart potentiality. Among the other 3-chloro-2-fluoro derivatives, **8e** with 4-methylsulfonylphenyl at oxadiazole ring displayed good inhibition with MIC value of $3.125 \mu\text{g mL}^{-1}$ against *S. aureus* and *E. faecalis*, but exhibited moderate activity against Gram-negative bacterial strains. While the other compounds **8b**, **8c** and **8d** did not exert good inhibition against all bacterial strains except compound **8c** with MIC value of $6.25 \mu\text{g mL}^{-1}$ against *E. coli*. Among the 2,4-dichlorophenyl derivatives, compound **8f** showed good inhibition against *E. faecalis* with MIC value of $3.125 \mu\text{g mL}^{-1}$ and $6.25 \mu\text{g mL}^{-1}$ against *S. aureus*, and also compound **8j** having methylsulfonyl group showed moderate inhibition against both Gram-positive bacterial strains with MIC value of $6.25 \mu\text{g mL}^{-1}$. However, the rest of the compounds did not show any pronounced inhibition. From the results of 3-chloro-2-fluoro derivatives, compound **8n** with 3-chlorobenzyl group at

Table 2 Antibacterial and antifungal activity data of the target compounds **8a–t** in terms of MIC^a

Sample	Antibacterial activity (MIC in $\mu\text{g mL}^{-1}$)				Antifungal activity (MIC in $\mu\text{g mL}^{-1}$)	
	Gram positive		Gram negative			
	<i>S. aureus</i>	<i>E. faecalis</i>	<i>E. coli</i>	<i>K. pneumoniae</i>	<i>C. albicans</i>	<i>A. fumigates</i>
Control	—	—	—	—	—	—
8a	3.125	1.6	3.125	3.125	1.6	3.125
8b	25.0	50.0	12.5	25.0	12.5	25.0
8c	25.0	25.0	6.25	25.0	12.5	25.0
8d	50.0	25.0	50.0	50.0	12.5	12.5
8e	3.125	3.125	12.5	12.5	6.25	3.125
8f	6.25	3.125	25.0	12.5	3.125	6.25
8g	25.0	50.0	12.5	25.0	25	12.5
8h	12.5	25.0	25.0	12.5	6.25	12.5
8i	12.5	6.25	25.0	50.0	12.5	6.25
8j	6.25	12.5	6.25	12.5	3.125	3.125
8k	1.6	1.6	3.125	6.25	0.8	1.6
8l	25.0	12.5	50.0	50.0	25	25
8m	12.5	12.5	12.5	12.5	3.125	3.125
8n	1.6	3.125	3.125	12.5	1.6	1.6
8o	6.25	12.5	12.5	12.5	25	25.0
8p	6.25	3.125	12.5	6.25	1.6	1.6
8q	3.125	3.125	6.25	12.5	3.125	1.6
8r	3.125	1.6	3.125	6.25	3.125	6.25
8s	6.25	6.25	3.125	3.125	1.6	1.6
8t	25.0	12.5	50.0	50.0	12.5	6.25
Ciprofloxacin	2.0	1.0	2.0	1.0	—	—
Fluconazole	—	—	—	—	16.0	8.0

^a Bold font indicates a significant activity of compounds against the respective bacterial strains.

oxadiazole ring showed good inhibition against *S. aureus* with MIC value of $3.125 \mu\text{g mL}^{-1}$ and against *E. faecalis* with MIC of $6.25 \mu\text{g mL}^{-1}$. Among the compounds **8p**, **8q**, **8r**, **8s** and **8t**, all the compounds displayed good inhibition against Gram-positive bacterial strains, except compound **8t** with MIC $\geq 12.5 \mu\text{g mL}^{-1}$. Against *K. pneumoniae* and *E. coli*, it did not show any considerable activity. The high activity of the derivatives might be due to the presence of dihalo substitution at the phenyl ring beside the benzimidazole moiety.

2.2.2. Antifungal activity. *In vitro* antifungal activities of the synthesized compounds **8a–t** were assessed against two representative fungal pathogens, *C. albicans* and *A. fumigates*, by the broth dilution method using fluconazole as a standard drug. Results are presented in Table 2. Most of the compounds showed excellent inhibition with MIC values less than that of the standard drug fluconazole (MIC: $8.0 \mu\text{g mL}^{-1}$) against fungal pathogen *A. fumigates*, except compounds **8c**, **8d**, **8g** and **8j**. Interestingly, compounds **8k**, **8m** and **8n** with 3-chloro-2-fluorophenyl substitutions at the 2nd position of benzimidazole and chloro substitution at oxadiazole end showed minimum MIC value ($1.6 \mu\text{g mL}^{-1}$) with highest inhibition among the compounds in the series against *A. fumigates*. Against *C. albicans*, compound **8a** showed the highest inhibition with MIC value of $1.6 \mu\text{g mL}^{-1}$ and compounds **8b**, **8c**, **8d**, **8e**, **8f**, **8h**, **8i**, **8j**, **8k**, **8l**, **8m**, **8n**, **8o**, **8p** and **8r** showed more inhibition than the standard drug. The rest of the derivatives showed moderate activity. Presence of chlorofluoro substitution at benzimidazole end enhanced the activity.

2.2.3. Antimycobacterial study. Based on encouraging results from the antibacterial screening, the selected title compounds **8a**, **8e**, **8f**, **8j**, **8p**, **8n**, **8p**, **8q**, **8r** and **8s** which showed good antibacterial activity were further screened for their *in vitro* antimycobacterial activity against *Mycobacterium tuberculosis* H37Rv using pyrazinamide, streptomycin, ciprofloxacin and isoniazid as standards by Microplate Alamar Blue Assay (MABA) method.⁴² The corresponding results are summarized in Table 3. As evident from the table, all the tested compounds exhibited moderate to excellent anti-tubercular activity with MIC ranging from 1.6 to $25.0 \mu\text{g mL}^{-1}$. It is interesting to note that most of the tested compounds showed either enhanced activity or activity in line with the reference drugs used. Surprisingly, compounds **8k**, **8n**, **8p** and **8r** showed excellent *in vitro* activity against *M. tuberculosis* H37Rv with MIC value of $1.6 \mu\text{g mL}^{-1}$ which was approximately two fold times lower than reference drugs pyrazinamide and ciprofloxacin (MIC: $3.12 \mu\text{g mL}^{-1}$) and four fold time lower than streptomycin and isoniazid (MIC: $6.25 \mu\text{g mL}^{-1}$). Compounds **8k** and **8n** contain 3-chloro-2-fluorophenyl substitution at the 2nd position of benzimidazole and chloro substitution at the other extreme end aryl group attached to the 5th position of oxadiazole ring. Whereas, compounds **8p** and **8r** have 2,3-dichlorophenyl substitution at phenyl ring attached to benzimidazole moiety at the 2nd position, and *para*-chloro and *ortho*-chlorophenyl substitution respectively at oxadiazole ring end. Also, compounds **8a**, **8f** and **8s** showed potent activity with MIC value of $3.12 \mu\text{g mL}^{-1}$, which was equipotent with standard drugs pyrazinamide and ciprofloxacin (MIC: $3.12 \mu\text{g mL}^{-1}$) and about two fold times more

Table 3 Antimycobacterial activity of selected target molecules against *M. tuberculosis* H37Rv, HC₅₀ values for selected compounds by hemolytic test and selectivity ratio of potent molecules^a

No.	Compound	MIC ($\mu\text{g mL}^{-1}$)	HC ₅₀ ($\mu\text{g mL}^{-1}$)	Selectivity ratio (HC ₅₀ /MIC)
	Control	—	—	—
1	8a	3.125	226.72	72.55
2	8e	25.0	ND	—
3	8f	3.125	ND	—
4	8j	25.0	ND	—
5	8k	1.6	874.29	546.30
6	8n	1.6	93.69	58.55
7	8p	1.6	129.46	80.91
8	8q	12.5	ND	—
9	8r	1.6	369.05	230.63
10	8s	3.125	280.84	89.86
11	Isoniazid	6.25	ND	—
12	Pyrazinamide	3.125	ND	—
13	Streptomycin	6.25	1016.34	162.56
14	Ciprofloxacin	3.125	ND	—

^a ND = not determined.

active than streptomycin and isoniazid (MIC: $6.25 \mu\text{g mL}^{-1}$). All these highly active compounds possess 3-chloro-2-fluoro and 2,3-dichlorophenyl substitutions at benzimidazole end and monohaloaryl substitution at the 5th position of oxadiazole moiety. It has been reported that lipophilicity of compounds plays a major role in their biological activity.⁴³ This is found to be an important parameter which is related to membrane permeation in biological systems. In addition, halogen-substituted molecules have a prominent place in modern medicinal chemistry and they will improve the lipophilic nature of compounds.⁴⁴ The better activity of the compounds could be due to the presence of 2,3-dihalophenyl substitutions at the 2nd position of the benzimidazole ring which increases the lipophilic nature of the compound, thereby making the molecule more cell permeable. However, isosteric displacement of 4-chlorophenyl group in **8p** with 3-pyridinyl group in compound **8q** at phenyl ring attached to the 3rd position of oxadiazole ring showed a reduction in the activity with MIC value of $12.5 \mu\text{g mL}^{-1}$. Similarly, introduction of methylsulfonyl group (**8j**) in place of 4-chloro substitution (**8f**) showed significant loss in potential against *M. tuberculosis* M37Rv with MIC value of $25 \mu\text{g mL}^{-1}$. Further, when 4-chloro group of **8a** was replaced with methylsulfonyl group, activity against *M. tuberculosis* M37Rv was drastically reduced. In addition, it could also be observed that all the 4-chlorophenyl substituted compounds **8a**, **8f**, **8k** and **8p** at the 3rd position of oxadiazole ring, with 2-chloro-6-fluoro, 2,4-dichloro, 3-chloro-2-fluoro, and 2,3-dichlorophenyl substitutions at the 2nd position of benzimidazole showed potent *in vitro* activity against *M. tuberculosis* M37Rv (MIC $\leq 3.12 \mu\text{g mL}^{-1}$). Also, it could be noted that 2,3-dichloro (**8p**) and 3-chloro-2-fluoro (**8k**) derivatives showed higher activity than 2-chloro-6-fluoro (**8a**) and 2,4-dichloro (**8f**) derivatives.

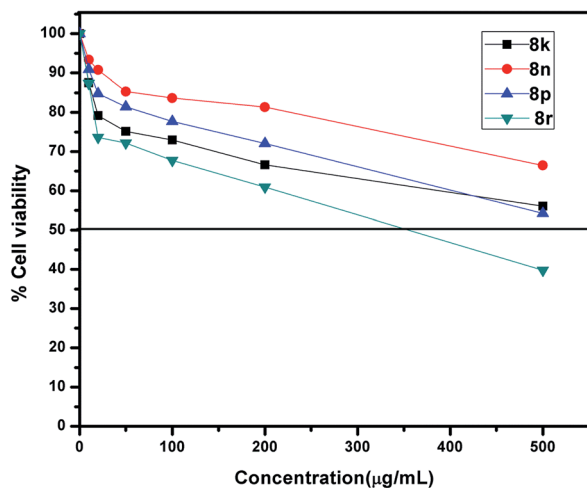
2.2.4. Hemolytic activity. The preliminary toxicity of selected compounds was investigated against human

erythrocytes (hRBCs)⁴⁵ and results are displayed in Table 3. The concentration corresponding to 50% cell lysis was expressed as HC_{50} value. The results indicated that the compounds having good anti-tubercular activity showed significant lysis only at higher concentration in the range of 93–874 $\mu\text{g mL}^{-1}$. The most potent compounds **8k** and **8r** remain nontoxic even at very high HC_{50} values of 874 and 369 $\mu\text{g mL}^{-1}$ respectively. Compounds **8n** and **8p** are also found to be nontoxic at higher concentration (HC_{50} value 93.69 and 129.46 $\mu\text{g mL}^{-1}$). The selectivity ratio

(HC_{50}/MIC) was also calculated for selected compounds. All the tested compounds showed good selectivity ratio (HC_{50}/MIC) ≥ 58 . Compound **8k** with 3-chloro-2-fluoro substitution emerged as the most effective compound, being 546-fold selective toward *M. tuberculosis* M37Rv, and compound **8r** was found to be 230-fold selective. The selectivity for these compounds was higher than that for the standard drug streptomycin (selectivity index: 162). The high selectivity of compounds clearly implies the suitability of these compounds for drug development as new anti-TB candidates.

2.2.5. Cytotoxicity test. In order to further investigate these compounds as selective antimicrobial agents, toxicity testing was performed against human normal embryonic kidney cell line HEK 293 using MTT assay⁴⁶ with the most potent compounds **8k**, **8n**, **8p** and **8r**. A dose dependent cytotoxic effect was obtained with an increase in the concentration of the selected compounds as shown in Fig. 2. IC_{50} concentrations are given in Table 4. As we can observe from the results, the IC_{50} value for compound **8r** was found to be 355.42 $\mu\text{g mL}^{-1}$ and the remaining compounds **8k**, **8n** and **8p** showed IC_{50} values of more than 500 $\mu\text{g mL}^{-1}$, which is many fold higher concentration than the MIC for anti-tubercular activity. In order to compare the effectiveness of the compounds, the toxicity test was performed with standard drug mitomycin C which showed IC_{50} value of less than 100 $\mu\text{g mL}^{-1}$. This clearly supports that these compounds are nontoxic even at very high concentration.

2.2.6. In silico physicochemical properties. The assessment of drug-likeness nature of active compounds in pharmaceutical research plays an important role in lead generation and lead optimization. Lipinski's "rule of five" helps to develop bioactive drug candidates.^{47,48} For a bioactive molecule to be a therapeutic agent, it should satisfy Lipinski's rule of five along with high oral bioavailability, which is decided by the ADME (absorption, distribution, metabolism and elimination) profile. According to this rule, an active compound should have molecular weight ≤ 500 , partition coefficient ($\log P$) should be ≤ 5 , and number of hydrogen bond donors should be ≤ 5 and number of hydrogen bond acceptors ≤ 10 . Along with this rule, the number of rotatable bonds in a molecule should be ≤ 10 and topological polar surface area (TPSA) should be $\leq 140 \text{ \AA}^2$ for an effective orally bioavailable compound. TPSA is the sum of surfaces of polar atoms such as oxygen, nitrogen and attached hydrogen and the number of rotatable bonds is a measure of



Values are the mean $\pm$ standard deviation of three independent experiments.

Fig. 2 Dose response curves of the cytotoxic effect of selected compounds on HEK 293 cell line. Values are the mean $\pm$ standard deviation of three independent experiments.

Table 4 Cytotoxicity test: IC_{50} values of the selected compounds against normal human embryonic cell line HEK 293

No.	Compound	IC_{50} value ($\mu\text{g mL}^{-1}$)
1	8k	>500
2	8n	>500
3	8p	>500
4	8r	355.42
5	Mitomycin C	<100

Table 5 In silico physicochemical properties of selected target compounds^a

Compound	MV	MW	milog P	nON	nOHNH	nROTB	%ABS	TPSA ($\text{\AA}^2$)	nVioltn
8a	345.29	439.28	6.70	5	0	3	89.42	56.75	1
8f	353.89	455.73	7.34	5	0	3	89.42	56.75	1
8k	345.29	439.28	6.80	5	0	3	89.42	56.75	1
8n	362.09	453.30	6.73	5	0	4	89.42	56.75	1
8p	353.89	455.73	7.32	5	0	3	89.42	56.75	1
8r	353.89	455.73	7.27	5	0	3	89.42	56.75	1
8s	347.08	469.75	7.30	5	0	4	89.42	56.75	1

^a MV: molecular volume; MW: molecular weight; milog P: logarithm of partition coefficient between *n*-octanol and water; nON: number of hydrogen bond donors; nOHNH: number of hydrogen bond donors; nROTB: number of rotatable bonds, TPSA: topological polar surface area; %ABS: percentage absorption; nVioltn: number of violations.

conformational flexibility of the molecule which in turn decides the binding of receptors. The percentage absorption was calculated by the equation $\%ABS = [109 - (0.345)TPSA]$.

In our present study, the physicochemical descriptors of the potent compounds **8a**, **8f**, **8k**, **8n**, **8p**, **8r** and **8s** were assessed to determine the ADME parameters by using the Molinspiration online property calculation toolkit and the results are summarized in Table 5. From the table we can observe that all the compounds showed excellent passive oral absorption with percentage absorption >90%. All the target compounds satisfy all the parameters except the log P value, which is slightly higher than the limiting value for all the compounds. But, violation of any one of the rules does not affect the bioavailability of the molecule and drug-likeness,⁴⁹ if the percentage absorption is good. TPSA was found to be 56.75 Å² for all the compounds and numbers of rotatable bonds were less than 10 for all the compounds which decide molecular flexibility.

3. Conclusion

In conclusion, this work is focused on the development of novel potent compounds based on the benzimidazole–oxadiazole compact system. A series of twenty novel compounds of benzimidazoles with [1,2,4]-oxadiazoles were designed and efficiently synthesized with good yield by employing rapid ‘one pot’ nitro reductive cyclization. Compounds **8a** and **8k** emerged as promising antimicrobial agents against all the bacterial and fungal pathogens tested and are found to be superior to ciprofloxacin and fluconazole with good ADME parameters. Compounds **8k** and **8n** with 3-chloro-2-fluoro aryl ring at the 2nd position of benzimidazole and 4-chlorophenyl (**8k**) and 3-chlorobenzyl (**8n**) substitution at oxadiazole ring emerged as excellent anti-tubercular agents with twofold more activity than pyrazinamide and ciprofloxacin, and four times more than streptomycin and isoniazid. Also, compounds **8p** and **8r** with 2,3-dichlorophenyl substitutions at benzimidazole end and 4-chloro (**8p**) and 2-chlorophenyl (**8r**) substitutions at oxadiazole phenyl ring are found to be around two times more active than pyrazinamide and ciprofloxacin, and fourfold more than streptomycin and isoniazid against *M. tuberculosis* M37Rv, and which emerged as potent anti-tuberculosis agents along with good drug-like properties. The preliminary toxicity testing of potent compounds **8a**, **8k**, **8n**, **8p**, **8r** and **8s** against human erythrocytes showed less toxicity and good selectivity index. Compound **8k** showed the highest selectivity index of 546-fold. Further investigation of cytotoxicity of potent compounds against human embryonic kidney cell line HEK 293 revealed the nontoxicity of these compounds. Structure–activity relationships revealed that the presence of dihalo substitution at the 2nd position aryl ring of benzimidazole moiety as well as halogen substitution at the phenyl ring of [1,2,4]-oxadiazole are crucial for pronounced activity and significantly enhanced antibacterial, antifungal and antimycobacterial activities. Further detailed structural modifications at 1st, 2nd and 5th positions of benzimidazole moiety with different heterocyclic rings are underway to get better analogs with improved efficacy.

4. Experimental section

4.1. Materials and methods

4.1.1. Preparation of *N*¹-hydroxyamidines (2). Sodium bicarbonate (70 mmol, 5.88 g) was added in portions to a solution of hydroxylamine hydrochloride (70 mmol, 4.79 g) in 25 mL of water. A solution of substituted acetonitrile (35 mmol) in 40 mL of ethanol was then added, and the mixture stirred under reflux for 8 h. The precipitate formed was filtered off and recrystallized from ethanol.

4.1.1.1. 4-Chloro-*N*¹-hydroxybenzamidine (2a). White crystalline solid; FT IR (ATR, $\nu_{\max}$, cm⁻¹): 3348 (N–H), 3207 (OH), 3030 (Ar–H), 2979, 2928 (C–H), 1652 (C=N); ¹H NMR (400 MHz, DMSO-*d*₆, δ ppm): 8.98 (s, OH), 7.78 (d, 2H, *p*-Cl phenyl ring, *J* = 8.8 Hz), 7.50 (d, 2H, *p*-Cl phenyl ring, *J* = 8.4 Hz), 5.45 (s, 2H, NH₂); ESI-MS: 171.1 (M⁺ + H), 173.1 [(M + H) + 2]⁺; anal. calcd for C₇H₇ClN₂O: C, 49.28; H, 4.14; N, 16.42. Found: C, 49.27; H, 4.15; N, 16.41.

4.1.1.2. *N*¹-Hydroxynicotinamidine (2b). Off white solid; FT IR (ATR, $\nu_{\max}$, cm⁻¹): 3350 (N–H), 3200 (OH), 3020 (Ar–H), 2989, 2933 (C–H), 1650 (C=N); ¹H NMR (400 MHz, DMSO-*d*₆, δ ppm): 5.49 (s, 2H, NH₂), 7.58 (d, 1H, Ar–H, *J* = 8.4 Hz), 8.03 (s, 1H, Ar–H), 8.16–8.20 (m, 2H, Ar–H), 8.98 (s, OH); ESI-MS: 138.14 (M⁺ + H); anal. calcd for C₆H₇N₃O: C, 52.55; H, 5.14; N, 30.64. Found: C, 52.54; H, 5.15; N, 30.65.

4.1.1.3. 2-Chloro-*N*¹-hydroxybenzamidine (2c). White crystalline solid; FT IR (ATR, $\nu_{\max}$, cm⁻¹): 3360 (N–H), 3220 (OH), 3025 (Ar–H), 2980, 2930 (C–H), 1649 (C=N); ¹H NMR (400 MHz, DMSO-*d*₆, δ ppm): 5.49 (s, 2H, NH₂), 7.26–7.37 (m, 2H, Ar–H), 7.42 (m, 1H, Ar–H), 7.52 (dd, 1H, Ar–H, *J* = 7.5 and 1.4 Hz), 8.98 (s, OH); ESI-MS: 171.6 (M + H)⁺, 173.3 [(M + H) + 2]⁺; anal. calcd for C₇H₇ClN₂O: C, 49.28; H, 4.14; N, 16.42. Found: C, 49.27; H, 4.15; N, 16.43.

4.1.1.4. 3-Chlorophenyl-*N*¹-hydroxyacetamidine (2d). White crystalline solid; FT IR (ATR, $\nu_{\max}$, cm⁻¹): 3366 (N–H), 3204 (OH), 3015 (Ar–H), 2985, 2937 (C–H), 1649 (C=N); ¹H NMR (400 MHz, DMSO-*d*₆, δ ppm): 3.32 (s, 2H, benzyl-CH₂), 7.93 (d, 1H, Ar–H, *J* = 8.8 Hz), 8.08 (t, 1H, Ar–H, *J* = 8.4 Hz), 8.26 (d, 1H, Ar–H, *J* = 8.4 Hz), 8.48 (s, 1H, Ar–H); ESI-MS: 185.06 (M⁺ + H), 187.4 [(M + H) + 2]⁺; anal. calcd for C₈H₉ClN₂O: C, 52.04; H, 4.91; N, 15.17. Found: C, 52.05; H, 4.90; N, 15.18.

4.1.1.5. *N*¹-Hydroxy-4-(methylsulfonyl)benzamidine (2e). Light brown crystals; FT IR (ATR, $\nu_{\max}$, cm⁻¹): 3414 (OH), 3263 (NH), 3081 (Ar–H), 2968 (C–H), 1653 (C=N), 1379 (asym SO₂), 1083 (sym SO₂); ¹H NMR (400 MHz, DMSO-*d*₆, δ ppm): 2.71 (s, 2H, –CH₃), 5.45 (s, 2H, NH₂), 7.47 (d, 2H, Ar–H, *J* = 8.4 Hz), 7.60 (d, 2H, Ar–H, *J* = 8 Hz), 8.93 (s, OH); ESI-MS: 214.5 (M⁺ + H), 215.1 [(M + H) + 1]⁺; anal. calcd for C₈H₁₀N₂O₃S: C, 44.85; H, 4.70; N, 13.08; S, 14.97. Found: C, 44.86; H, 4.71; N, 13.09; S, 14.98.

4.1.2. Preparation of ethyl 4-chloro-3-nitrobenzoate (4). A mixture of ethyl 4-chloro-3-nitrobenzoic acid (10 mmol) in 30 mL of ethanol and 12 mL of conc. hydrochloric acid was refluxed for 10 h. After completion of the reaction, solvent was removed under reduced pressure and the residue was taken up in ether (100 mL). The solution was extracted with saturated aqueous sodium bicarbonate solution (2 × 100 mL) and water

(2 × 100 mL). Reaction progress was confirmed by dichloromethane-methanol mixture (9 : 1).

Yellow solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3097 (Ar-H), 2976, 2941 (C-H), 1535 (NO_2 asym), 1354 (NO_2 sym); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, δ ppm): 1.34 (t, 3H, $-\text{CH}_3$, $J = 7.2$ Hz), 4.34–4.39 (q, 2H, ester CH_2), 7.94 (d, 1H, Ar-H), 8.18–8.21 (dd, 1H, Ar-H, $J = 8.4$ Hz and $J = 2$ Hz), 8.51 (s, 1H, Ar-H); ESI-MS: 232.0 $[(\text{M} + \text{H}) + 1]^+$, 234.1 $[(\text{M} + \text{H}) + 3]^+$; anal. calcd for $\text{C}_9\text{H}_8\text{ClNO}_4$: C, 47.08; H, 3.51; N, 15.44. Found: C, 47.07; H, 3.50; N, 15.43.

4.1.3. Preparation of ethyl 4-(methylamino)-3-nitrobenzoate (5). Ethyl 4-chloro-3-nitrobenzoate (4) (10 mmol, 4.5 g) was added to 25 mL of THF. To this were added methylamine (88 mmol, 2.7 g) and triethylamine (44 mmol, 4.4 g). Reaction mixture was kept stirring overnight with heating. After completion of the reaction, reaction mixture was quenched into ice water. Pure product was obtained which was collected by filtration.

Light yellow solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3369 (N-H), 3037 (Ar-H), 2926, 2883 (C-H), 1535 (NO_2 asym), 1354 (NO_2 sym); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, δ ppm): 1.36 (t, 3H, $-\text{CH}_3$, $J = 7.2$ Hz), 3.01 (s, 3H, $-\text{CH}_3$), 4.34–4.39 (q, 2H, ester CH_2), 6.75 (d, 1H, Ar-H, $J = 8.8$ Hz), 7.59–7.72 (m, 1H, Ar-H), 8.11 (s, 1H, Ar-H), 12.41 (s, NH); ESI-MS: 224.5 ($\text{M}^+ - \text{H}$); anal. calcd for $\text{C}_{10}\text{H}_{12}\text{N}_2\text{O}_4$: C, 53.57; H, 5.39; N, 12.49. Found: C, 53.58; H, 5.39; N, 12.48.

4.1.4. General procedure for the synthesis of ethyl 2-aryl-1-methyl-1H-benzo[d]imidazole-5-carboxylate (6). Sodium dithionite (40 mmol, 6.98 g) was added to a stirred solution of ethyl 4-(methylamino)-3-nitrobenzoate (5) (13 mmol, 2.12 g) and substituted benzaldehyde (13 mmol) in DMSO (20 mL). The reaction mixture was stirred at 90 °C for 3 h. After the completion of reaction (monitored by TLC with ethyl acetate : hexane 3 : 7, v/v), it was poured into crushed ice. The solid separated was filtered off, washed with water, and dried. Collected product was recrystallized from dimethylformamide.

4.1.4.1. Ethyl 2-(2-chloro-6-fluorophenyl)-1-methyl-1H-benzo[d]imidazole-5-carboxylate (6a). White powder; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3061 (Ar-H), 2972, 2926 (C-H), 1699 ($\text{C}=\text{O}$), 1620 ($\text{C}=\text{N}$), 1284 (C-F), 896 (C-Cl); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, δ ppm): 1.36 (t, 3H, $J = 7.2$ Hz, $-\text{CH}_3$), 3.70 (s, 3H, $-\text{CH}_3$), 4.33–4.39 (q, 2H, $-\text{CH}_2-\text{CH}_3$, $J = 14.4$ and 6.8 Hz), 7.52 (t, 1H, Ar-H, $J = 8.4$ Hz), 7.62 (d, 1H, Ar-H, $J = 7.2$ Hz), 7.71–7.77 (m, 1H, Ar-H), 7.80 (d, 1H, Ar-H, $J = 8.4$ Hz), 7.98–8.01 (dd, 1H, Ar-H, $J = 8.8$ and 1.6 Hz), 8.34 (s, 1H, Ar-H); ESI-MS (m/z): 333.1 ($\text{M}^+ + \text{H}$), 335.1 $[(\text{M} + \text{H}) + 2]^+$; anal. calcd for $\text{C}_{17}\text{H}_{14}\text{ClFN}_2\text{O}_2$: C, 61.36; H, 4.24; N, 8.42. Found: C, 61.32; H, 4.25; N, 8.46.

4.1.4.2. Ethyl 2-(2,4-dichlorophenyl)-1-methyl-1H-benzo[d]imidazole-5-carboxylate (6b). White amorphous solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3040 (Ar-H), 2983, 2932 (C-H), 1692 ($\text{C}=\text{O}$), 1614 ($\text{C}=\text{N}$), 1538 ($\text{C}=\text{C}$), 890 (C-Cl); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, δ ppm): 1.38 (t, 3H, $J = 7.2$ Hz, $-\text{CH}_3$), 3.73 (s, 3H, $-\text{CH}_3$), 4.34–4.40 (q, 2H, $-\text{CH}_2-\text{CH}_3$, $J = 14.4$ and 6.8 Hz), 7.58 (s, 1H, Ar-H), 7.66 (d, 1H, Ar-H, $J = 7.2$ Hz), 7.98–8.01 (dd, 1H, Ar-H, $J = 8.8$ and 1.6 Hz), 8.09 (d, 2H, Ar-H, $J = 8.8$ Hz), 8.37 (s, 1H, Ar-H); ESI-MS (m/z): 349.1 ($\text{M}^+ + \text{H}$), 351.6 $[(\text{M} + \text{H}) + 2]^+$, 353.4 $[(\text{M} + \text{H}) + 4]^+$; anal. calcd for $\text{C}_{17}\text{H}_{14}\text{Cl}_2\text{N}_2\text{O}_2$: C, 58.47; H, 4.04; N, 8.02. Found: C, 58.48; H, 4.07; N, 8.02.

4.1.4.3. Ethyl 2-(3-chloro-2-fluorophenyl)-1-methyl-1H-benzo[d]imidazole-5-carboxylate (6c). White amorphous solid; FT IR

(ATR, $\nu_{\max}$, cm^{-1}): 3045 (Ar-H), 2981, 2931 (C-H), 1699 ($\text{C}=\text{O}$), 1609 ($\text{C}=\text{N}$), 1539 ($\text{C}=\text{C}$), 1272 (C-F), 890 (C-Cl); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, δ ppm): 1.36 (t, 3H, 7.2 Hz, $-\text{CH}_3$), 3.73 (s, 3H, $-\text{CH}_3$), 4.32–4.39 (q, 2H, $-\text{CH}_2-\text{CH}_3$, $J = 14.4$ and 6.8 Hz), 7.42–7.47 (m, 3H, 3-Cl-2-F phenyl ring), 7.69 (d, 1H, Ar-H, $J = 7.2$ Hz), 7.99–8.02 (dd, 1H, Ar-H, $J = 8.8$ and 1.6 Hz), 8.38 (s, 1H, Ar-H); ESI-MS (m/z): 333.1 ($\text{M}^+ + \text{H}$), 335.7 $[(\text{M} + \text{H}) + 2]^+$; anal. calcd for $\text{C}_{17}\text{H}_{14}\text{ClFN}_2\text{O}_2$: C, 61.36; H, 4.24; N, 8.42. Found: C, 61.38; H, 4.22; N, 8.43.

4.1.4.4. Ethyl 2-(2,3-dichlorophenyl)-1-methyl-1H-benzo[d]imidazole-5-carboxylate (6d). White crystalline solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3049 (Ar-H), 2990, 2939 (C-H), 1700 ($\text{C}=\text{O}$), 1623 ($\text{C}=\text{N}$), 1539 ($\text{C}=\text{C}$), 889 (C-Cl); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, δ ppm): 1.38 (t, 3H, 7.2 Hz, $-\text{CH}_3$), 3.75 (s, 3H, $-\text{CH}_3$), 4.31–4.39 (q, 2H, $-\text{CH}_2-\text{CH}_3$, $J = 14.4$ and 6.8 Hz), 7.60 (d, 1H, Ar-H, $J = 7.2$ Hz), 7.68 (d, 1H, Ar-H, $J = 8.4$ Hz), 7.83 (t, 1H, Ar-H, $J = 8.4$ Hz), 7.99–8.02 (dd, 1H, Ar-H, $J = 8.8$ and 1.6 Hz), 8.09 (d, 1H, Ar-H, $J = 8.8$ Hz), 8.38 (s, 1H, Ar-H); ESI-MS (m/z): 350.1 ($\text{M}^+ + \text{H}$), 352.6 $[(\text{M} + \text{H}) + 2]^+$, 354.2 $[(\text{M} + \text{H}) + 4]^+$; anal. calcd for $\text{C}_{17}\text{H}_{14}\text{Cl}_2\text{N}_2\text{O}_2$: C, 58.47; H, 4.04; N, 8.02. Found: C, 58.45; H, 4.05; N, 8.04.

4.1.5. Procedure for the preparation of 2-aryl-1-methyl-1H-benzo[d]imidazole-5-carboxylic acid (7). To a solution of ethyl-2-aryl-1-methyl-1H-benzo[d]imidazole-5-carboxylate (6) (9 mmol, 3 g) in 20 mL of ethanol, sodium hydroxide dissolved in the minimum quantity of water was added slowly with stirring. After complete addition, the reaction mixture was refluxed at 80 °C for 5 h. After completion of the reaction, reaction mixture was quenched into ice water. The pure product was collected by filtration.

4.1.5.1. 2-(2-Chloro-6-fluorophenyl)-1-methyl-1H-benzo[d]imidazole-5-carboxylic acid (7a). Creamy white powder; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3196 (OH), 3030 (Ar-H), 2970, 2924 (C-H), 1689 ($\text{C}=\text{O}$), 1618 ($\text{C}=\text{N}$), 1566 ($\text{C}=\text{C}$), 1033 (C-F), 752 (C-Cl); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, δ ppm): 3.69 (s, 1H, $-\text{CH}_3$), 7.52 (t, 1H, Ar-H, $J = 8.8$ Hz), 7.61 (d, 1H, Ar-H, $J = 8$ Hz), 7.71–7.78 (m, 2H, Ar-H), 7.98 (d, 1H, Ar-H, $J = 8.4$ Hz), 8.30 (s, 1H, Ar-H), 12.82 (bs, OH, 1H); ESI-MS (m/z): 305.0 ($\text{M}^+ + \text{H}$), 307.1 $[(\text{M} + \text{H}) + 2]^+$. Anal. calcd for $\text{C}_{15}\text{H}_{10}\text{ClFN}_2\text{O}_2$: C, 59.13; H, 3.31; N, 9.19. Found: C, 59.16; H, 3.32; N, 9.22.

4.1.5.2. 2-(2,4-Dichlorophenyl)-1-methyl-1H-benzo[d]imidazole-5-carboxylic acid (7b). White amorphous solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3240 (OH), 2984, 2939 (C-H), 1684 ($\text{C}=\text{O}$), 1616 ($\text{C}=\text{N}$), 1528 ($\text{C}=\text{C}$), 895 (C-Cl); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, δ ppm): 3.73 (s, 3H, $-\text{CH}_3$), 7.59 (s, 1H, Ar-H), 7.66 (d, 1H, Ar-H, $J = 7.2$ Hz), 7.99–8.02 (dd, 1H, Ar-H, $J = 8.8$ and 1.6 Hz), 8.09 (d, 2H, Ar-H, $J = 8.8$ Hz), 8.37 (s, 1H, Ar-H), 12.80 (bs, OH, 1H); ESI-MS (m/z): 320.1 ($\text{M}^+ - \text{H}$), 322.4 ($\text{M} + \text{H}$) $^+$; anal. calcd for $\text{C}_{15}\text{H}_{10}\text{Cl}_2\text{N}_2\text{O}_2$: C, 56.10; H, 3.14; N, 8.72. Found: C, 56.08; H, 3.17; N, 8.75.

4.1.5.3. 2-(3-Chloro-2-fluorophenyl)-1-methyl-1H-benzo[d]imidazole-5-carboxylic acid (7c). Light cream solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3194 (OH), 2981, 2931 (C-H), 1689 ($\text{C}=\text{O}$), 1599 ($\text{C}=\text{N}$), 1539 ($\text{C}=\text{C}$), 1279 (C-F), 892 (C-Cl); ^1H NMR (400 MHz, $\text{DMSO}-d_6$, δ ppm): 3.73 (s, 3H, $-\text{CH}_3$), 7.43–7.48 (m, 3H, 3-Cl-2-F phenyl ring), 7.69 (d, 1H, Ar-H, $J = 7.2$ Hz), 7.99–8.02 (dd, 1H, Ar-H, $J = 8.8$ and 1.6 Hz), 8.38 (s, 1H, Ar-H), 12.73 (bs, OH, 1H); ESI-MS (m/z): 305.7 ($\text{M}^+ + \text{H}$),

307.2 $[(M + H) + 2]^+$; anal. calcd for $C_{15}H_{10}ClFN_2O_2$: C, 59.13; H, 3.31; N, 9.19. Found: C, 59.10; H, 3.34; N, 9.18.

4.1.5.4. 2-(2,3-Dichlorophenyl)-1-methyl-1H-benzo[d]imidazole-5-carboxylic acid (7d). White amorphous solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3149 (OH), 2990, 2939 (C–H), 1690 (C=O), 1603 (C=N), 1549 (C=C), 886 (C–Cl); ^1H NMR (400 MHz, DMSO- d_6 , δ ppm): 3.75 (s, 3H, $-\text{CH}_3$), 7.60 (d, 1H, Ar–H, $J = 7.2$ Hz), 7.68 (d, 1H, Ar–H, $J = 8.4$ Hz), 7.83 (t, 1H, Ar–H, $J = 8.4$ Hz), 7.99–8.02 (dd, 1H, Ar–H, $J = 8.8$ and 1.6 Hz), 8.09 (d, 1H, Ar–H, $J = 8.8$ Hz), 8.38 (s, 1H, Ar–H), 12.43 (bs, OH, 1H); ESI-MS (m/z): 322.1 ($M^+ + H$), 324.3 $[(M + H) + 2]^+$, 326.3 $[(M + H) + 4]^+$; anal. calcd for $C_{15}H_{10}Cl_2N_2O_2$: C, 56.10; H, 3.14; N, 8.72. Found: C, 56.14; H, 3.15; N, 8.74.

4.1.6. General procedure for the preparation of 2-aryl-5-(3-(4-chlorophenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzo[d]imidazole (8a–t). A mixture of 2-aryl-1-methyl-1H-benzo[d]imidazole-5-carboxylic acid (7) (10 mmol, 2.84 g), TBTU (12.5 mmol, 4 g) and NMM (15 mmol, 1.51 g) in DMF solvent was kept stirred under cold condition. After one hour of stirring, N^1 -hydroxyamidine (2) was added to the reaction mixture in portions and allowed to stir overnight at room temperature. After completion of the reaction, the separated solid was collected by filtration and washed well with water. Obtained product was dried well.

4.1.6.1. 2-(2-Chloro-6-fluorophenyl)-5-(3-(4-chlorophenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzo[d]imidazole (8a). Pale yellow solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3018 (Ar–H), 2976, 2945 (C–H), 1618 (C=N), 1596 (C=C), 1165 (C–F), 754 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.67 (s, 3H, $-\text{CH}_3$), 7.34 (d, 2H, Ar–H, $J = 7.6$ Hz), 7.42–7.46 (m, 3H, 2-Cl-6-F phenyl ring), 7.52 (d, 2H, *p*-Cl phenyl ring, $J = 8.4$ Hz), 8.08 (d, 2H, *p*-Cl phenyl ring, $J = 8.8$ Hz), 8.17–8.19 (dd, 1H, $J = 8.4$ Hz and 1.2 Hz), 8.65 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.72 ($-\text{CH}_3$), 110.56, 114.60, 118.63, 121.17, 123.31, 125.78, 128.85, 132.63, 132.70, 137.18, 138.44, 143.06, 160.43, 162.44 ($-\text{C}=\text{N}$), 168.13 ($-\text{C}=\text{N}$), 176.53 ($-\text{C}=\text{N}$); ESI-MS: 439.2 ($M^+ + H$), 441.2 $[(M + H) + 2]^+$, 443.2 $[(M + H) + 4]^+$; anal. calcd for $C_{22}H_{13}Cl_2FN_4O$: C, 60.15; H, 2.98; N, 12.75. Found: C, 60.18; H, 2.99; N, 12.79.

4.1.6.2. 2-(2-Chloro-6-fluorophenyl)-5-(3-(pyridin-3-yl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzo[d]imidazole (8b). Yellow solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3028 (Ar–H), 2996, 2935 (C–H), 1608 (C=N), 1596 (C=C), 1145 (C–F), 784 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.69 (s, 3H, $-\text{CH}_3$), 7.43–7.47 (m, 3H, 2-Cl-6-F phenyl ring), 7.54 (d, 1H, Ar–H, $J = 8.4$ Hz), 7.58 (d, 1H, Ar–H, $J = 8.4$ Hz), 8.08 (s, 1H, Ar–H), 8.16–8.20 (m, 2H, Ar–H), 8.24–8.27 (dd, 1H, benzimidazole-H, $J = 8.4$ Hz and 1.2 Hz), 8.69 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 31.72 ($-\text{CH}_3$), 111.50, 114.65, 119.64, 120.17, 122.61, 124.88, 129.87, 131.64, 132.75, 136.18, 139.44, 144.06, 148.37, 159.43, 161.47 ($-\text{C}=\text{N}$), 169.13 ($-\text{C}=\text{N}$), 177.53 ($-\text{C}=\text{N}$); ESI-MS: 407.3 ($M^+ + H$), 409.3 $[(M + H) + 2]^+$; anal. calcd for $C_{21}H_{13}ClFN_5O$: C, 62.15; H, 3.23; N, 17.26. Found: C, 62.16; H, 3.26; N, 17.28.

4.1.6.3. 2-(2-Chloro-6-fluorophenyl)-5-(3-(2-chlorophenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzo[d]imidazole (8c). White solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3054 (Ar–H), 2986, 2935 (C–H), 1618 (C=N), 1586 (C=C), 1148 (C–F), 786 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.64 (s, 3H, $-\text{CH}_3$), 7.34–7.38 (m, 3H, Ar–H, $J = 8$ Hz), 7.49 (d, 1H, Ar–H, $J = 8.4$ Hz), 7.53–7.56 (m, 4H, Ar–H), 8.20–8.22 (dd, 1H, benzimidazole-H, $J = 8.8$ Hz and 1.6 Hz), 8.60 (s, 1H,

benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 32.32 ($-\text{CH}_3$), 113.55, 115.65, 120.64, 122.17, 124.65, 126.88, 127.33, 129.87, 132.64, 133.75, 136.58, 138.49, 144.06, 157.43, 160.47 ($-\text{C}=\text{N}$), 166.13 ($-\text{C}=\text{N}$), 177.53 ($-\text{C}=\text{N}$); ESI-MS: 440.2 ($M^+ + H$), 442.4 $[(M + H) + 2]^+$, 444.7 $[(M + H) + 4]^+$; anal. calcd for $C_{22}H_{13}Cl_2FN_4O$: C, 60.15; H, 2.98; N, 12.75. Found: C, 60.17; H, 2.96; N, 12.78.

4.1.6.4. 2-(2-Chloro-6-fluorophenyl)-5-(3-(3-chlorobenzyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzo[d]imidazole (8d). White amorphous solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3069 (Ar–H), 2976, 2929 (C–H), 1602 (C=N), 1576 (C=C), 1188 (C–F), 796 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.30 (s, 2H, benzyl- CH_2), 3.71 (s, 3H, imidazole- CH_3), 7.47–7.54 (m, 3H, 2-Cl-6-F phenyl ring-H), 7.59 (d, 1H, Ar–H, $J = 8$ Hz), 7.70–7.73 (m, 1H, Ar–H, benzimidazole-H), 7.76 (d, 1H, Ar–H, $J = 8.4$ Hz, benzimidazole-H), 7.90 (d, 1H, Ar–H, $J = 8.8$ Hz), 8.08 (t, 1H, Ar–H, $J = 8.4$ Hz), 8.43 (s, 1H, Ar–H), 8.57 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.99 ($-\text{CH}_3$), 36.46 (CH_2), 110.79, 121.59, 124.24, 125.97, 126.64, 127.47, 127.82, 128.56, 128.52, 130.16, 146.92, 157.59, 163.92 ($-\text{C}=\text{N}$), 169.53 ($-\text{C}=\text{N}$), 175.86 ($-\text{C}=\text{N}$); ESI-MS: 453.2 ($M^+ + H$), 455.2 $[(M + H) + 2]^+$, 457.2 $[(M + H) + 4]^+$; anal. calcd for $C_{23}H_{15}Cl_2FN_4O$: C, 60.94; H, 3.34; N, 12.36. Found: C, 60.95; H, 3.35; N, 12.38.

4.1.6.5. 2-(2-Chloro-6-fluorophenyl)-5-(3-(4-(methylsulfonyl)-phenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzo[d]imidazole (8e). White amorphous solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3099 (Ar–H), 2996, 2939 (C–H), 1613 (C=N), 1571 (C=C), 1388 (asym SO_2), 1185 (C–F), 1129 (sym SO_2), 798 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.24 (s, 3H, CH_3), 3.63 (s, 3H, CH_3), 7.66–7.75 (m, 5H, Ar–H), 7.96 (d, 2H, Ar–H, $J = 8.8$ Hz), 8.12 (d, 2H, Ar–H, $J = 8.4$ Hz), 8.57 (s, 1H, Ar–H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 31.72 ($-\text{CH}_3$), 111.56, 116.60, 119.63, 120.17, 126.73, 129.75, 133.63, 134.70, 136.62, 137.11, 139.49, 148.06, 161.43, 163.44 ($-\text{C}=\text{N}$), 169.13 ($-\text{C}=\text{N}$), 176.53 ($-\text{C}=\text{N}$); ESI-MS: 483.2 ($M^+ + H$), 484.2 $[(M + H) + 2]^+$; anal. calcd for $C_{23}H_{16}ClFN_4O_3S$: C, 57.20; H, 3.34; N, 11.60; S, 6.64. Found: C, 57.21; H, 3.37; N, 11.63; S, 6.63.

4.1.6.6. 2-(2,4-Dichlorophenyl)-5-(3-(4-chlorophenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzo[d]imidazole (8f). Light yellow solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3077 (Ar–H), 2992, 2930 (C–H), 1623 (C=N), 1571 (C=C), 778 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.68 (s, 3H, $-\text{CH}_3$), 7.39 (d, 1H, benzimidazole-H, $J = 8$ Hz), 7.43 (d, 2H, 4-chlorophenyl, $J = 8.4$ Hz), 7.50 (d, 2H, 4-chlorophenyl, $J = 8.4$ Hz), 7.53 (s, 1H, 2,4-dichlorophenyl), 8.08 (d, 2H, 2,4-dichlorophenyl, $J = 8.8$ Hz), 8.16–8.19 (dd, 1H, benzimidazole-H, $J = 8.4$ Hz and 1.6 Hz), 8.62 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 31.20 ($-\text{CH}_3$), 111.55, 119.75, 121.97, 123.36, 125.62, 127.74, 128.86, 129.15, 129.88, 133.18, 137.22, 152.60 ($-\text{C}=\text{N}$), 168.16 ($-\text{C}=\text{N}$), 176.51 ($-\text{C}=\text{N}$); ESI-MS: 455.1 ($M^+ + H$), 457.1 $[(M + H) + 2]^+$, 459.1 $[(M + H) + 4]^+$, 461.0 $[(M + H) + 6]^+$; anal. calcd for $C_{23}H_{15}Cl_2FN_4O$: C, 57.98; H, 2.88; N, 12.29. Found: C, 57.96; H, 2.86; N, 12.30.

4.1.6.7. 2-(2,4-Dichlorophenyl)-5-(3-(pyridin-3-yl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzo[d]imidazole (8g). Light brown solid; FT IR (ATR, $\nu_{\max}$, cm^{-1}): 3057 (Ar–H), 2994, 2932 (C–H), 1629 (C=N), 1576 (C=C), 778 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.68 (s, 3H, $-\text{CH}_3$), 7.40 (d, 2H, Ar–H, $J = 8.4$ Hz), 7.56 (s, 1H, Ar–H), 7.59 (d, 1H, Ar–H, $J = 8$ Hz), 7.63 (d, 1H, Ar–H, $J = 7.6$ Hz), 8.08–8.12 (m, 3H, Ar–H), 8.17–8.20 (dd, 1H,

benzimidazole-H, $J = 8.4$ Hz and 1.2 Hz), 8.64 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.20 ($-\text{CH}_3$), 109.55 , 116.57 , 118.24 , 121.98 , 122.36 , 124.62 , 125.41 , 127.74 , 128.36 , 129.45 , 129.88 , 130.31 , 133.18 , 137.22 , 138.20 , 154.64 ($-\text{C}=\text{N}$), 169.16 ($-\text{C}=\text{N}$), 175.59 ($-\text{C}=\text{N}$); ESI-MS: 455.1 ($\text{M}^+ + \text{H}$), 457.5 $[(\text{M} + \text{H}) + 2]^+$; anal. calcd for $\text{C}_{21}\text{H}_{13}\text{Cl}_2\text{N}_5\text{O}$: C, 59.73 ; H, 3.10 ; N, 16.79 . Found: C, 59.75 ; H, 3.11 ; N, 16.82 .

4.1.6.8. *2-(2,4-Dichlorophenyl)-5-(3-(2-chlorophenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (8h)*. White amorphous solid; FT IR (ATR, ν_{max} , cm^{-1}): 3057 (Ar-H), 2994 , 2932 (C-H), 1629 ($\text{C}=\text{N}$), 1576 ($\text{C}=\text{C}$), 778 (C-Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.64 (s, 3H, $-\text{CH}_3$), 7.34 (d, 2H, Ar-H, $J = 8.4$ Hz), 7.52 – 7.55 (m, 4H, Ar-H), 7.77 (d, 1H, benzimidazole-H, $J = 8$ Hz), 7.87 (s, 1H, Ar-H), 8.22 – 8.25 (dd, 1H, $J = 8.4$ Hz and 1.2 Hz), 8.65 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.29 ($-\text{CH}_3$), 110.59 , 118.45 , 120.94 , 124.36 , 125.82 , 127.76 , 128.80 , 129.85 , 129.99 , 133.18 , 135.32 , 136.22 , 137.46 , 140.39 , 142.36 , 153.60 ($-\text{C}=\text{N}$), 166.76 ($-\text{C}=\text{N}$), 174.71 ($-\text{C}=\text{N}$); ESI-MS: 455.1 ($\text{M}^+ + \text{H}$), 457.1 $[(\text{M} + \text{H}) + 2]^+$, 459.5 $[(\text{M} + \text{H}) + 4]^+$, 461.0 $[(\text{M} + \text{H}) + 6]^+$; anal. calcd for $\text{C}_{22}\text{H}_{13}\text{Cl}_3\text{N}_4\text{O}$: C, 57.98 ; H, 2.88 ; N, 12.29 . Found: C, 57.99 ; H, 2.89 ; N, 12.30 .

4.1.6.9. *2-(2,4-Dichlorophenyl)-5-(3-(3-chlorobenzyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (8i)*. White solid; FT IR (ATR, ν_{max} , cm^{-1}): 3059 (Ar-H), 2973 , 2928 (C-H), 1629 ($\text{C}=\text{N}$), 1570 ($\text{C}=\text{C}$), 783 (C-Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.32 (s, 2H, benzyl- CH_2), 3.75 (s, 3H, imidazole- CH_3), 7.47 (d, 2H, 2,4-dichlorophenyl-H, $J = 8.4$ Hz), 7.59 (s, 1H, 2,4-dichlorophenyl-H), 7.76 (d, 1H, benzimidazole-H, $J = 8.4$ Hz), 7.80 – 7.83 (m, 1H, Ar-H, benzimidazole-H), 7.90 (d, 1H, Ar-H, $J = 8.8$ Hz), 8.08 (t, 1H, Ar-H, $J = 8.4$ Hz), 8.16 (d, 1H, Ar-H, $J = 8.4$ Hz), 8.43 (s, 1H, Ar-H), 8.57 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.82 ($-\text{CH}_3$), 35.39 ($-\text{CH}_2$), 111.59 , 119.45 , 121.94 , 124.48 , 126.82 , 128.76 , 129.80 , 129.85 , 130.99 , 132.46 , 133.28 , 134.18 , 136.45 , 138.43 , 142.40 , 144.20 , 153.68 ($-\text{C}=\text{N}$), 166.77 ($-\text{C}=\text{N}$), 174.77 ($-\text{C}=\text{N}$); ESI-MS: 471.4 ($\text{M}^+ + \text{H}$), 473.1 $[(\text{M} + \text{H}) + 2]^+$, 475.4 $[(\text{M} + \text{H}) + 4]^+$, 477.0 $[(\text{M} + \text{H}) + 6]^+$; anal. calcd for $\text{C}_{23}\text{H}_{15}\text{Cl}_3\text{N}_4\text{O}$: C, 58.81 ; H, 3.22 ; N, 11.93 . Found: C, 58.83 ; H, 3.23 ; N, 11.95 .

4.1.6.10. *2-(2,4-Dichlorophenyl)-5-(3-(4-(methylsulfonyl)phenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (8j)*. Light cream solid; FT IR (ATR, ν_{max} , cm^{-1}): 3081 (Ar-H), 2971 , 2935 (C-H), 1382 (asym SO_2), 1123 (sym SO_2); ^1H NMR (400 MHz, DMSO, δ ppm): 3.20 (s, 3H, CH_3), 3.68 (s, 3H, CH_3), 7.66 – 7.75 (m, 5H, Ar-H), 7.91 (d, 2H, Ar-H, $J = 8.8$ Hz), 8.09 (d, 2H, Ar-H, $J = 8.4$ Hz), 8.57 (s, 1H, Ar-H). ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.91 (CH_3), 43.60 (CH_2), 110.69 , 121.45 , 123.49 , 124.14 , 127.06 , 127.86 , 128.22 , 129.42 , 129.65 , 133.72 , 134.25 , 136.00 , 138.60 , 139.25 , 141.90 , 143.18 , 151.67 ($-\text{C}=\text{N}$), 157.31 ($-\text{C}=\text{N}$), 163.95 ($-\text{C}=\text{N}$); ESI-MS: 513.2 ($\text{M}^+ + \text{H}$), 515.2 $[(\text{M} + \text{H}) + 2]^+$, 417.1 $[(\text{M} + \text{H}) + 4]^+$; anal. calcd for $\text{C}_{23}\text{H}_{16}\text{Cl}_2\text{N}_4\text{O}_3\text{S}$: C, 55.32 ; H, 3.23 ; N, 11.22 ; S, 6.42 . Found: C, 55.33 ; H, 3.24 ; N, 11.24 ; S, 6.43 .

4.1.6.11. *2-(3-Chloro-2-fluorophenyl)-5-(3-(4-chlorophenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (8k)*. White crystalline solid; FT IR (ATR, ν_{max} , cm^{-1}): 3034 (Ar-H), 2970 , 2924 (C-H), 1618 ($\text{C}=\text{N}$), 1566 ($\text{C}=\text{C}$), 1133 (C-F), 782 (C-Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.64 (s, 3H, $-\text{CH}_3$), 7.35 (d, 2H, Ar-H, $J = 7.6$ Hz), 7.41 – 7.45 (m, 3H, 3-Cl-2-F phenyl ring), 7.55 (d, 1H, p -Cl phenyl ring, $J = 8.4$ Hz), 8.08 (d, 2H, p -Cl phenyl ring, $J =$

8.8 Hz), 8.18 – 8.20 (dd, 1H, $J = 8.4$ Hz and 1.2 Hz), 8.65 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.47 , 111.64 , 118.74 , 119.86 , 122.61 , 123.48 , 124.40 , 126.97 , 128.57 , 131.56 , 132.26 , 138.05 , 142.09 , 143.04 , 155.89 ($-\text{C}=\text{N}$), 156.73 ($-\text{C}=\text{N}$), 162.86 ($-\text{C}=\text{N}$); ESI-MS: 440.3 ($\text{M}^+ + \text{H}$), 441.2 $[(\text{M} + \text{H}) + 2]^+$, 443.2 $[(\text{M} + \text{H}) + 4]^+$; anal. calcd for $\text{C}_{22}\text{H}_{13}\text{Cl}_2\text{FN}_4\text{O}$: C, 60.15 ; H, 2.98 ; N, 12.75 . Found: C, 60.17 ; H, 2.97 ; N, 12.78 .

4.1.6.12. *2-(3-Chloro-2-fluorophenyl)-5-(3-(pyridin-3-yl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (8l)*. Light cream solid; FT IR (ATR, ν_{max} , cm^{-1}): 3039 (Ar-H), 2966 , 2935 (C-H), 1608 ($\text{C}=\text{N}$), 1596 ($\text{C}=\text{C}$), 1145 (C-F), 784 (C-Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.69 (s, 3H, $-\text{CH}_3$), 7.42 – 7.46 (m, 3H, 3-Cl-2-F phenyl ring), 7.55 (d, 1H, Ar-H, $J = 8.4$ Hz), 7.62 (d, 1H, Ar-H, $J = 8.4$ Hz), 8.08 (s, 1H, Ar-H), 8.16 – 8.20 (m, 2H, Ar-H), 8.70 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.87 (CH_3), 112.63 , 117.72 , 118.86 , 123.61 , 124.48 , 124.98 , 126.97 , 128.47 , 129.53 , 130.56 , 132.66 , 134.72 , 138.55 , 141.49 , 143.04 , 144.25 , 155.89 ($-\text{C}=\text{N}$), 156.73 ($-\text{C}=\text{N}$), 162.86 ($-\text{C}=\text{N}$); ESI-MS: 407.3 ($\text{M}^+ + \text{H}$), 409.3 $[(\text{M} + \text{H}) + 2]^+$; anal. calcd for $\text{C}_{21}\text{H}_{13}\text{ClFN}_5\text{O}$: C, 62.15 ; H, 3.23 ; N, 17.26 . Found: C, 62.17 ; H, 3.22 ; N, 17.29 .

4.1.6.13. *2-(3-Chloro-2-fluorophenyl)-5-(3-(2-chlorophenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (8m)*. Creamy white solid; FT IR (ATR, ν_{max} , cm^{-1}): 3059 (Ar-H), 2986 , 2932 (C-H), 1628 ($\text{C}=\text{N}$), 1583 ($\text{C}=\text{C}$), 1149 (C-F), 787 (C-Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.64 (s, 3H, $-\text{CH}_3$), 7.35 – 7.39 (m, 3H, 3-Cl-2-F phenyl ring, $J = 8$ Hz), 7.50 (d, 1H, Ar-H, $J = 8.4$ Hz), 7.53 – 7.56 (m, 4H, Ar-H), 8.21 – 8.23 (dd, 1H, $J = 8.8$ Hz and 1.6 Hz), 8.65 (s, 1H, benzimidazole-H); ^1H NMR (400 MHz, DMSO, δ ppm): 3.80 (s, 3H, $-\text{CH}_3$), 7.19 – 7.87 (m, 9H, Ar-H), 8.71 (s, Ar-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 31.27 (CH_3), 110.84 , 119.44 , 119.56 , 121.67 , 123.46 , 124.40 , 125.93 , 127.03 , 129.57 , 131.22 , 132.16 , 132.26 , 132.88 , 139.05 , 142.09 , 154.04 , 155.89 ($-\text{C}=\text{N}$), 156.03 ($-\text{C}=\text{N}$), 163.81 ($-\text{C}=\text{N}$); ESI-MS: 440.6 ($\text{M}^+ + \text{H}$), 442.4 $[(\text{M} + \text{H}) + 2]^+$, 444.7 $[(\text{M} + \text{H}) + 4]^+$; anal. calcd for $\text{C}_{22}\text{H}_{13}\text{Cl}_2\text{FN}_4\text{O}$: C, 60.15 ; H, 2.98 ; N, 12.75 . Found: C, 60.17 ; H, 2.97 ; N, 12.76 .

4.1.6.14. *2-(3-Chloro-2-fluorophenyl)-5-(3-(3-chlorobenzyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (8n)*. White amorphous solid; FT IR (ATR, ν_{max} , cm^{-1}): 3059 (Ar-H), 2973 , 2925 (C-H), 1602 ($\text{C}=\text{N}$), 1586 ($\text{C}=\text{C}$), 1184 (C-F), 798 (C-Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.33 (s, 2H, benzyl- CH_2), 3.72 (s, 3H, imidazole- CH_3), 7.48 – 7.55 (m, 3H, 3-Cl-2-F phenyl ring-H), 7.60 (d, 1H, Ar-H, $J = 8$ Hz), 7.70 – 7.73 (m, 1H, Ar-H, benzimidazole-H), 7.78 (d, 1H, Ar-H, $J = 8.4$ Hz, benzimidazole-H), 7.90 (d, 1H, Ar-H, $J = 8.8$ Hz), 8.08 (t, 1H, Ar-H, $J = 8.4$ Hz), 8.43 (s, 1H, Ar-H), 8.61 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 31.77 (CH_3), 42.42 (CH_2), 110.84 , 118.64 , 119.45 , 120.52 , 122.61 , 124.46 , 125.40 , 126.93 , 127.03 , 130.57 , 131.82 , 132.16 , 132.86 , 132.98 , 139.65 , 142.59 , 154.34 , 155.89 ($-\text{C}=\text{N}$), 157.03 ($-\text{C}=\text{N}$), 162.81 ($-\text{C}=\text{N}$); ESI-MS: 454.4 ($\text{M}^+ + \text{H}$), 455.2 $[(\text{M} + \text{H}) + 2]^+$, 457.2 $[(\text{M} + \text{H}) + 4]^+$; anal. calcd for $\text{C}_{23}\text{H}_{15}\text{Cl}_2\text{FN}_4\text{O}$: C, 60.91 ; H, 3.34 ; N, 12.36 . Found: C, 60.93 ; H, 3.36 ; N, 12.33 .

4.1.6.15. *2-(3-Chloro-2-fluorophenyl)-5-(3-(4-(methylsulfonyl)phenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (8o)*. FT IR (ATR, ν_{max} , cm^{-1}): 3029 (Ar-H), 2986 , 2929 (C-H), 1618 ($\text{C}=\text{N}$), 1561 ($\text{C}=\text{C}$), 1386 (asym SO_2), 1188 (C-F), 1128 (sym SO_2), 794

(C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.26 (s, 3H, CH_3), 3.60 (s, 3H, CH_3), 7.66–7.75 (m, 5H, Ar–H), 7.96 (d, 2H, Ar–H, $J = 8.8$ Hz), 8.12 (d, 2H, Ar–H, $J = 8.4$ Hz), 8.57 (s, 1H, Ar–H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.73 (CH_3), 44.49 (CH_3), 112.84, 118.68, 120.59, 122.51, 124.49, 125.70, 126.93, 128.03, 130.27, 131.87, 132.06, 132.86, 134.98, 139.62, 143.51, 154.34, 155.89 (–C=N), 159.03 (–C=N), 161.48 (–C=N); ESI-MS: 483.2 ($\text{M}^+ + \text{H}$), 484.2 $[(\text{M} + \text{H}) + 2]^+$; anal. calcd for $\text{C}_{23}\text{H}_{16}\text{ClFN}_4\text{O}_3\text{S}$: C, 57.20; H, 3.34; N, 11.60; S, 6.64. Found: C, 57.21; H, 3.35; N, 11.61; S, 6.64.

4.1.6.16. 2-(2,3-Dichlorophenyl)-5-(3-(4-chlorophenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (**8p**). Light yellow solid; FT IR (ATR, ν_{max} , cm^{-1}): 3077 (Ar–H), 2992, 2930 (C–H), 1623 (C=N), 1571 (C=C), 778 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.68 (s, 3H, $-\text{CH}_3$), 7.42 (d, 1H, benzimidazole-H, $J = 8$ Hz), 7.45 (d, 2H, 4-chlorophenyl, $J = 8.4$ Hz), 7.50 (d, 2H, 4-chlorophenyl, $J = 8.4$ Hz), 7.68 (d, 1H, 2,3-dichlorophenyl, $J = 8.4$ Hz), 7.83 (t, 1H, 2,3-dichlorophenyl, $J = 8.4$ Hz), 8.09 (d, 1H, 2,3-dichlorophenyl, $J = 8.8$ Hz), 8.15–8.18 (dd, 1H, benzimidazole-H, $J = 8.4$ Hz and 1.6 Hz), 8.66 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 31.30 ($-\text{CH}_3$), 112.55, 119.75, 121.77, 123.36, 125.61, 127.84, 128.86, 129.15, 129.88, 133.18, 137.22, 154.60 (–C=N), 169.16 (–C=N), 176.81 (–C=N); ESI-MS: 455.5 ($\text{M}^+ + \text{H}$), 457.1 $[(\text{M} + \text{H}) + 2]^+$, 459.1 $[(\text{M} + \text{H}) + 4]^+$, 461.0 $[(\text{M} + \text{H}) + 6]^+$; anal. calcd for $\text{C}_{23}\text{H}_{15}\text{Cl}_2\text{FN}_4\text{O}$: C, 57.98; H, 2.88; N, 12.29. Found: C, 57.97; H, 2.89; N, 12.31.

4.1.6.17. 2-(2,3-Dichlorophenyl)-5-(3-(pyridin-3-yl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (**8q**). White solid; FT IR (ATR, ν_{max} , cm^{-1}): 3055 (Ar–H), 2993, 2927 (C–H), 1629 (C=N), 1579 (C=C), 773 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.69 (s, 3H, $-\text{CH}_3$), 7.40 (d, 1H, 2,3-dichlorophenyl, $J = 8.4$ Hz), 7.50 (t, 1H, 2,3-dichlorophenyl, $J = 8.4$ Hz), 7.55 (d, 1H, 2,3-dichlorophenyl, $J = 8.4$ Hz), 7.59 (d, 1H, Ar–H, $J = 8$ Hz), 7.63 (d, 1H, Ar–H, $J = 7.6$ Hz), 8.08–8.12 (m, 3H, Ar–H), 8.17–8.20 (dd, 1H, benzimidazole-H, $J = 8.4$ Hz and 1.2 Hz), 8.64 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.20 ($-\text{CH}_3$), 109.99, 116.57, 118.24, 121.98, 122.36, 124.62, 125.41, 127.74, 128.36, 129.45, 129.88, 130.31, 133.18, 137.22, 138.20, 154.64 (–C=N), 169.16 (–C=N), 175.59 (–C=N); ESI-MS: 455.1 ($\text{M}^+ + \text{H}$), 457.5 $[(\text{M} + \text{H}) + 2]^+$; anal. calcd for $\text{C}_{21}\text{H}_{13}\text{Cl}_2\text{N}_5\text{O}$: C, 59.73; H, 3.10; N, 16.79. Found: C, 59.72; H, 3.09; N, 16.80.

4.1.6.18. 2-(2,3-Dichlorophenyl)-5-(3-(2-chlorophenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (**8r**). White solid; FT IR (ATR, ν_{max} , cm^{-1}): 3057 (Ar–H), 2993, 2930 (C–H), 1621 (C=N), 1572 (C=C), 779 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.64 (s, 3H, $-\text{CH}_3$), 7.34 (d, 1H, 2,3-dichlorophenyl, $J = 8.4$ Hz), 7.52–7.55 (m, 4H, Ar–H), 7.77 (d, 1H, benzimidazole-H, $J = 8$ Hz), 7.87 (t, 1H, 2,3-dichlorophenyl, $J = 8.4$ Hz), 8.04 (d, 1H, 2,3-dichlorophenyl-H, $J = 8.6$ Hz), 8.22–8.25 (dd, 1H, $J = 8.4$ Hz and 1.2 Hz), 8.65 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.56 ($-\text{CH}_3$), 111.59, 117.45, 120.94, 124.36, 125.82, 126.76, 128.80, 129.85, 129.99, 133.18, 135.32, 136.22, 137.46, 140.39, 142.36, 153.70 (–C=N), 166.76 (–C=N), 134.71 (–C=N); ESI-MS: 455.1 ($\text{M}^+ + \text{H}$), 457.1 $[(\text{M} + \text{H}) + 2]^+$, 459.5 $[(\text{M} + \text{H}) + 4]^+$, 461.0 $[(\text{M} + \text{H}) + 6]^+$; anal. calcd for $\text{C}_{22}\text{H}_{13}\text{Cl}_3\text{N}_4\text{O}$: C, 57.98; H, 2.88; N, 12.29. Found: C, 57.99; H, 2.89; N, 12.28.

4.1.6.19. 2-(2,3-Dichlorophenyl)-5-(3-(3-chlorobenzyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (**8s**). White solid;

FT IR (ATR, ν_{max} , cm^{-1}): 3059 (Ar–H), 2973, 2928 (C–H), 1629 (C=N), 1575 (C=C), 814 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.32 (s, 2H, benzyl- CH_2), 3.75 (s, 3H, $-\text{CH}_3$), 7.47 (d, 1H, dichlorophenyl, $J = 8$ Hz), 7.59 (t, 1H, 2,3-dichlorophenyl, $J = 8$ Hz), 7.76 (d, 1H, benzimidazole-H, $J = 8.4$ Hz), 7.80 (d, 1H, dichlorophenyl, $J = 8$ Hz), 7.85–7.88 (m, 1H, Ar–H, benzimidazole-H), 7.96 (d, 1H, Ar–H, $J = 8.8$ Hz), 8.08 (t, 1H, Ar–H, $J = 8.4$ Hz), 8.16 (d, 1H, Ar–H, $J = 8.4$ Hz), 8.43 (s, 1H, Ar–H), 8.57 (s, 1H, benzimidazole-H); ^{13}C NMR (125 MHz, DMSO, δ ppm): 32.82 ($-\text{CH}_3$), 39.39 ($-\text{CH}_2$), 111.59, 119.45, 121.44, 124.48, 126.72, 129.76, 129.80, 129.85, 131.99, 132.46, 133.28, 134.18, 137.45, 138.43, 142.46, 144.54, 153.69 (–C=N), 166.77 (–C=N), 175.77 (–C=N); ESI-MS: 471.4, 473.1 $[(\text{M} + \text{H}) + 2]^+$, 475.4 $[(\text{M} + \text{H}) + 4]^+$, 477.0 $[(\text{M} + \text{H}) + 6]^+$; anal. calcd for $\text{C}_{23}\text{H}_{15}\text{Cl}_3\text{N}_4\text{O}$: C, 58.81; H, 3.22; N, 11.93. Found: C, 58.83; H, 3.24; N, 11.92.

4.1.6.20. 2-(2,3-Dichlorophenyl)-5-(3-(4-(methylsulfonyl)phenyl)-[1,2,4]-oxadiazol-5-yl)-1-methyl-1H-benzodimidazole (**8t**). Light orange solid; FT IR (ATR, ν_{max} , cm^{-1}): 3071 (Ar–H), 2971, 2935 (C–H), 1386 (asym SO_2), 1113 (sym SO_2), 803 (C–Cl); ^1H NMR (400 MHz, DMSO, δ ppm): 3.24 (s, 3H, $-\text{CH}_3$), 3.69 (s, 3H, $-\text{CH}_3$), 7.68–7.77 (m, 5H, Ar–H), 7.98 (d, 2H, Ar–H, $J = 8.8$ Hz), 8.18 (d, 2H, Ar–H, $J = 8.4$ Hz), 8.60 (s, 1H, Ar–H). ^{13}C NMR (125 MHz, DMSO, δ ppm): 30.95 (CH_3), 43.63 (CH_2), 111.69, 122.45, 123.49, 124.14, 127.06, 127.86, 128.22, 129.45, 129.64, 133.71, 134.25, 136.04, 138.67, 139.25, 141.90, 144.18, 152.67 (–C=N), 158.31 (–C=N), 164.95 (–C=N); ESI-MS: 513.1 ($\text{M}^+ + \text{H}$), 515.5 $[(\text{M} + \text{H}) + 2]^+$, 417.4 $[(\text{M} + \text{H}) + 4]^+$; anal. calcd for $\text{C}_{23}\text{H}_{16}\text{Cl}_2\text{N}_4\text{O}_3\text{S}$: C, 55.32; H, 3.23; N, 11.22; S, 6.42. Found: C, 55.34; H, 3.25; N, 11.24; S, 6.44.

4.2. Antimicrobial activity

The MIC of the samples in DMSO was determined by using different concentrations of compounds in brain heart infusion broth for bacteria and fungi by the macro dilution method. The lowest concentration of the sample in DMSO inhibiting the visible growth of microorganisms was considered as MIC. Here the broth dilution test was used to determine MIC of samples **8a–t**. Freshly prepared nutrient broth was used as diluent. The 24 h old cultures of the test bacteria *S. aureus*, *E. faecalis*, *E. coli* and *K. pneumoniae* and the test fungi *C. albicans* and *A. fumigatus* were diluted 100-fold in nutrient broth (NB) (100 μL bacterial cultures in 10 mL NB). A stock solution of the synthesized compounds with concentration of 1 mg mL^{-1} was prepared in DMSO. Different concentrations of the test samples (0.2, 0.4, 0.8, 1.6, 3.125, 6.25, 12.5, 25, 50 and 100 $\mu\text{g mL}^{-1}$) were added to the test tubes containing the bacterial and fungal cultures. All the tubes were incubated at 37 $^\circ\text{C}$ for 24 h for bacteria and at 28 $^\circ\text{C}$ for 48 h for fungi. The tubes were examined for visible turbidity and using NB as a control. Control without test samples and with solvent (DMSO) was assayed simultaneously. The lowest concentration that inhibited visible growth of the tested organisms was recorded as MIC.

4.3. In vitro anti-tubercular activity

Anti-tubercular activity was evaluated against *Mycobacterium tuberculosis* H37Rv by Microplate Alamar Blue Assay (MABA). In the presence of cellular metabolism, the colour of Alamar Blue

changes from a fully oxidized, non-fluorescent blue to a fully reduced, fluorescent pink colour. This methodology is nontoxic, uses a thermally stable reagent and shows good correlation with the BACTEC radiometric method. Middlebrook 7H9 broth base with Middlebrook OADC growth supplement was used as medium. 200 μL of sterile deionized water was added to all outer perimeter wells of sterile 96-well plate to minimize evaporation of medium in the test wells during incubation. The 96-well plate received 100 μL of the Middlebrook 7H9 broth. 10 mg of each compound was dissolved in 10.0 mL of DMSO separately and was further diluted with DMSO to make 100 $\mu\text{g mL}^{-1}$ and 10 $\mu\text{g mL}^{-1}$ stock solution. Using the above stock solution, serial dilutions of 0.2 $\mu\text{g mL}^{-1}$, 0.8 $\mu\text{g mL}^{-1}$, 1.6 $\mu\text{g mL}^{-1}$, 3.12 $\mu\text{g mL}^{-1}$, 6.25 $\mu\text{g mL}^{-1}$, 12.5 $\mu\text{g mL}^{-1}$, 25 $\mu\text{g mL}^{-1}$, 50 $\mu\text{g mL}^{-1}$ and 100 $\mu\text{g mL}^{-1}$ were made with DMSO directly on the plate. Further the positive control was prepared by inoculating the test organism with medium containing standard drugs and a negative control was prepared by inoculating test organism with medium containing only DMSO. 50 μL of DMSO was used as control. Medium without test compound/DMSO was also inoculated with the test organism to check whether the media will support the growth of the tubercle bacilli or not, and no mycobacteria growth was observed. The abovementioned wells were inoculated with *M. tuberculosis* H37Rv strain. Plates were covered and sealed with Parafilm and incubated at 37 °C for five days. After this time, 25 μL of freshly prepared 1 : 1 mixture of Alamar Blue reagent and 10% Tween 80 was added to the plate and incubated for 24 h. A blue color in the well was interpreted as no bacterial growth, and pink color was scored as growth.

4.4. Hemolytic assay

Human blood was collected in an EDTA (2 mg mL⁻¹)-containing vacutainer. Plasma and buffy coat were removed from the EDTA-blood by centrifugation at 800 $\times g$ for 10 min. Later, the erythrocytes were washed three times by using normal saline (0.9%) and resuspended in saline to 5% erythrocytes suspension. The cells were incubated for 1 h at 37 °C in the presence of test compounds. After the incubation, the samples were centrifuged at 800 $\times g$ for 10 min and the absorbance was read using a UV spectrophotometer. 2% Triton X-100 (Sigma-Aldrich, St Louis, USA) served as positive control. Phosphate buffered saline served as negative control. The absorbance of haemoglobin release was measured at 540 nm and is expressed as % of Triton X-100 induced hemolysis. The result was calculated by using the following formula:

$$\% \text{ Hemolysis} = \frac{[(\text{absorbance of sample}) - (\text{absorbance of blank})]}{[\text{highest absorbance of positive control}]} \times 100$$

Each concentration had triplicate values. HC₅₀ was determined for each triplicate absorbance measurement and average of HC₅₀ was plotted against concentration fitted with a sigmoid plot. From the curve, concentration corresponding to 50% hemolysis was determined.

4.5. In vitro cytotoxic activity

Human embryonic kidney cell line HEK 293 was procured from NCCS, Pune. The cytotoxicity effects of **8k**, **8n**, **8p** and **8r** were studied by MTT assay. The cells were cultured using Eagle's minimum essential medium (MEM) supplemented with 10% fetal bovine serum, penicillin (100 U mL⁻¹) and streptomycin (100 $\mu\text{g mL}^{-1}$) and grown in 25 cm² tissue culture flasks at 37 °C with 5% CO₂ incubator (Galaxy 170S, New Brunswick). After attaining 70–80% confluence, cells were trypsinized and seeded to 96-well plates in 100 μL of MEM (5000 cells/well). After 24 h, different concentrations of drugs were dissolved in DMSO and added to 96-well plates and incubated for 24 h at standard condition. Untreated cells represented a control group. Four replicates were set up for each test group. After completion of the incubation time, 20 μL of MTT dye (5 mg mL⁻¹ in PBS) was added to all wells; the plate was sealed with aluminium foil and incubated for 4 h in an incubator. After 4 h, 100 μL of DMSO was added to all the wells and mixed by uniform shaking. Using a multi-plate reader (LisaScan®EM) the absorbance was recorded at 570 nm. The percentage of viable cells was calculated using the following formula:

$$\text{Percentage of viability cells} = \frac{(\text{test sample-blank}/\text{control-blank})}{\times 100}$$

Each concentration had triplicate values and the average value was plotted against concentration by fitting with a sigmoid plot. From the curve, the concentration corresponding to 50% cell viability was noted.

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